SABER: Support-Conditioned Adaptive Bellman Error Reweighting for Sparse HER Actor-Critic

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Abstract

Hindsight Experience Replay (HER)는 sparse goal-conditioned reinforcement learning에서 실패 trajectory를 relabeled Bellman targets로 바꾸지만, 그 residual들이 current-policy action 근방의 critic fitting에 얼마나 유용한지는 결정하지 않는다. 본 논문은 이 차이를 hindsight Bellman critic-fit usefulness gap으로 제기한다.

SABER (Support-Conditioned Adaptive Bellman Error Reweighting)는 current policy action의 local replay-action support를 이용해 sample-wise weight를 만들고, 이를 critic Bellman residual에만 적용한다. HER relabeling, replay sampling, actor objective, backbone target construction, and critic architecture는 유지된다. Support predicate는 target noise의 직접 측정치가 아니라 critic fitting pressure를 배정하기 위한 local-support proxy다.

FetchPickAndPlace에서 SABER는 one-update-per-step DDPG/TD3/SAC HER baselines보다 mean final success와 mean AUC가 높고, TD3/SAC AUC paired intervals는 0을 제외한다. DrawerOpen DDPG는 HER 대비 large-effect case를 제공한다. Random/PER controls, DDPG/TD3/SAC diagnostics, high replay-ratio HER, REDQ-HER, and an HGR-style sampler-only ablation은 효과가 task/backbone dependent이며 pure support-reliability mechanism으로 환원되지 않음을 보여준다. 특히 compute/backbone-scaled references는 SABER의 environment-step AUC를 match하거나 능가할 수 있다. 따라서 SABER는 SOTA 또는 wall-clock dominance claim이 아니라, standard matched HER actor-critic에서 post-HER critic-loss allocation을 분리하는 lightweight objective로 해석한다.

Introduction

Goal-conditioned reinforcement learning은 agent가 하나의 고정된 reward가 아니라 다양한 goal을 달성하도록 학습하는 문제를 다룬다. 특히 robotic manipulation에서는 pushing, sliding, pick-and-place처럼 성공 조건은 명확하지만 reward가 sparse한 task가 많다. 이러한 setting에서는 대부분의 trajectory가 실패로 끝나기 때문에, naive off-policy actor-critic은 critic이 충분한 positive signal을 얻기 어렵다.

Hindsight Experience Replay (HER)는 이 문제를 해결하는 대표적인 방법이다 (Andrychowicz et al. 2017). HER는 agent가 원래 목표를 달성하지 못했더라도, 실제로 도달한 상태를 새로운 goal로 relabel하여 실패 경험을 학습 가능한 transition으로 바꾼다. 이 방식은 sparse reward에서도 goal-conditioned Bellman target을 대량으로 만들 수 있게 해준다.

그러나 HER가 해결하지 않는 문제가 있다. HER는 relabeled goal을 만들어 Bellman target의 quantity를 늘리지만, 그 update가 현재 policy improvement에 얼마나 유용한 critic supervision인지는 판단하지 않는다. Off-policy actor-critic에서 critic은 replay action을 중심으로 Bellman regression을 수행하고, actor는 같은 state-goal input에서 critic이 높게 평가하는 current policy action을 따라간다. 이때 현재 policy action이 local replay action support 바깥에 있다면, actor가 exploit하는 action region에 대한 critic estimate는 replay support가 있는 경우보다 더 취약할 수 있다. HER로 relabel된 transition이라고 해서 이 문제가 자동으로 사라지지는 않는다.

본 논문의 핵심 관찰은 다음이다.

> HER creates useful training goals, but it does not guarantee that every relabeled Bellman update is equally useful as critic supervision for the current policy.

우리는 이 문제를 hindsight Bellman critic-fit usefulness gap으로 부른다. 같은 HER batch 안에서도 어떤 sample은 local state-goal-horizon context에서 현재 policy action이 replay buffer에 support되어 있고, 어떤 sample은 그렇지 않다. 본 논문은 이 support predicate가 Bellman target noise나 sample-level target quality를 직접 측정한다고 주장하지 않는다. 대신 이를 current policy가 곧 query할 action region에 대한 local-support proxy로 사용한다. 즉, actor가 곧 query할 action region이 local replay support를 갖는 sample의 Bellman residual을 critic fitting에서 더 크게 반영한다.

이를 위해 우리는 SABER를 제안한다. SABER는 각 training sample에 대해 local replay-action support를 계산하고, support-positive sample의 critic Bellman error를 더 크게 반영하는 support-conditioned critic objective다. 중요한 점은 SABER가 HER를 대체하지 않는다는 것이다. HER는 training goal을 만들고, SABER는 그 goal-conditioned Bellman update 중 critic loss에서 더 강하게 반영할 update를 구분한다.

SABER의 main variant는 critic-only이다. 즉, support-derived weight는 critic TD loss에만 적용되며 actor objective와 replay sampling distribution은 baseline과 동일하게 유지된다. 이 설계는 SABER를 HER relabeling, prioritized replay, actor objective shaping과 분리하고, support signal

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의 효과를 critic Bellman fitting 쪽으로 좁혀 해석할 수 있게 한다.

따라서 본 논문은 architecture-scaled SOTA robotics benchmark가 아니라, standard one-update-per-step HER actor-critic 아래의 matched-backbone critic-loss allocation study다. High replay-ratio HER와 REDQ-HER rows는 compute/backbone-scaling references로 포함하지만, SABER가 REDQ-HER나 goal-selection/prioritization 계열을 압도한다는 claim은 하지 않는다. 더 정확히 말해 SABER는 one-update matched HER baselines를 개선하는 critic-side allocation intervention을 isolate하고, strong random-control 및 compute/backbone-scaling stress tests 아래에서도 이 intervention이 trivial fixed-count rule로 환원되지 않음을 보이는 데 초점을 둔다.

이 관점에서 SABER의 주된 가치는 최고 성능을 내는 별도의 RL algorithm이라는 데 있지 않다. 더 중요한 점은 HER 이후의 Bellman residual을 모두 같은 강도로 fitting하지 않고, current-policy action region과 더 연결된 update에 critic fitting pressure를 더 배정하는 좁고 모듈형인 intervention point를 제공한다는 것이다. 이 intervention은 HER goal generation, replay distribution, actor objective, critic architecture를 유지하므로 effect attribution이 상대적으로 깨끗하다. 또한 random controls, PER-style controls, high replay-ratio controls, and sample-level diagnostics를 함께 제시하면, support-conditioned weighting이 언제 generic reweighting이나 update scaling과 구분되고 언제 구분되지 않는지를 분석할 수 있다. 따라서 SABER는 strong compute-scaled methods의 replacement라기보다, sparse HER actor-critic에서 critic-side allocation을 실험하고 진단하기 위한 lightweight and interpretable layer로 위치시킨다.

본 논문의 contribution은 다음과 같다.

- Sparse-reward HER에서 relabeled target quantity와 current-policy action region의 critic-fit usefulness를 구분하는 hindsight Bellman critic-fit usefulness gap을 제기한다.
- Local replay-action support를 proxy로 사용해 critic Bellman residual만 재가중하고 HER relabeling, replay sampling, actor objective, and backbone target construction을 그대로 두는 SABER objective를 제안한다.
- FPP DDPG/TD3/SAC breadth, DrawerOpen large-effect evidence, random/PER controls, high-RR/REDQ scaling references, an HGR-style sampler-only control, and sample-level diagnostics를 함께 보고해 SABER의 효과와 mixed-mechanism boundary를 규정한다.

Related Work

Goal-Conditioned RL and HER

Goal-conditioned RL은 policy와 value function이 상태뿐 아니라 goal을 조건으로 받도록 학습한다. Sparse reward manipulation에서는 성공 signal이 드물기 때문에 goal relabeling이 특히 중요하다. HER는 실패 trajectory에서 실제로 달성된 state를 hindsight goal로 사용하여 sparse reward 문제를 완화한다 (Andrychowicz et al. 2017). HER의 핵심 역할은 training target을 늘리는 것이다.

SABER는 HER와 경쟁하는 방법이 아니다. HER가 "어떤 goal로 Bellman target을 만들 것인가"를 다룬다면, SABER는 "그렇게 만들어진 Bellman error를 critic이 얼마나 강하게 fitting할 것인가"를 다룬다. 따라서 SABER는 HER의 relabeling distribution을 바꾸는 방법이 아니라, HER 이후 critic loss weighting을 조정하는 방법이다.

Hindsight Goal Selection and Prioritization

HER 이후 많은 연구는 sparse goal-conditioned RL의 sample efficiency를 높이기 위해 어떤 hindsight goal, episode, transition을 더 자주 학습할지 바꾼다. HGG는 단기적으로 달성 가능하면서 장기 goal로 가는 데 유용한 hindsight goals를 생성한다 (Ren et al. 2019). CHER는 exploration-exploitation trade-off를 조정하는 curriculum-guided hindsight goal selection으로 제안되었다 (Fang et al. 2019). EBP는 trajectory energy를 기준으로 hindsight episodes를 우선순위화한다 (Zhao and Tresp 2018). Canonical HGR은 episode 평균 TD-error와 future hindsight goal의 TD-error를 사용해 두 단계 sampling probability를 바꾸고, 그 non-uniform sampling을 보정하는 episode/goal importance weights를 critic gradient에 적용한다 (Luu and Yoo 2021). MHER는 one-step HER target 대신 n-step relabeled returns를 사용하고, MHER(λ)와 model-based variants로 off-policy n-step bias를 줄인다 (Yang et al. 2021).

이 계열과 SABER의 차이는 intervention point에 있다. 위 방법들은 주로 relabeled goal distribution, replay episode/transition sampling, 또는 Bellman target construction을 바꾼다. SABER는 future-goal HER ratio와 uniform replay sampling을 유지하고, one-step HER Bellman target도 유지한다. 바뀌는 것은 critic Bellman residual의 sample-wise loss weight뿐이다.

HER bias를 직접 다루는 weighting methods도 SABER와 구분해야 한다. ARCHER는 real reward와 hindsight reward의 scale을 다르게 두어 HER bias를 완화한다 (Lanka and Wu 2018). USHER는 stochastic transition에서 HER Bellman estimator의 bias를 줄이기 위해 learned future-goal distribution으로 importance ratio를 구성해 Bellman MSE를 재가중한다 (Schramm et al. 2023). SABER weight는 bias-correcting likelihood ratio나 reward rescaling이 아니라 current-policy action의 local replay support로부터 얻는 heuristic allocation weight이며, unbiasedness를 주장하지 않는다.

Off-Policy Actor-Critic and Critic Error

DDPG, TD3, SAC와 같은 off-policy actor-critic 방법은 continuous control에서 널리 사용된다 (Lillicrap et al. 2015; Fujimoto, van Hoof, and Meger 2018; Haarnoja et al. 2018). 이 방법들은 replay buffer를 사용해 sample efficiency를 높이지만, critic approximation error와 actor exploitation 문제가 함께 발생할 수 있다. TD3는 twin critics, target policy smoothing, delayed policy update를 통해 overestimation과 function approximation error를 줄인다. SAC는 entropy-regularized objective를 통해 exploration과 robustness를 높인다.

High replay-ratio actor-critic도 중요한 비교 축이다. REDQ는 high update-to-data ratio, Q ensemble, random subset target minimization을 결합해 model-free continuous control의 sample efficiency를 높인다 (Chen et al. 2021). 최근 sparse goal-conditioned setting에서는 REDQ를 HER와 결합하고 target Q bounding을 추가해 Robotics tasks에서 강한 sample efficiency를 보인 work도 있다 (Hiraoka

Table 1: Closest HER-related methods by intervention point. SABER is positioned as critic-side Bellman residual reweighting rather than hindsight goal selection or replay prioritization.

Method	Main intervention	Main signal	Changes target or sampling?	Difference from SABER
HER	relabel failed transitions with achieved goals	future achieved goals	changes training goal distribution	SABER operates after HER relabeling
HGG / CHER	hindsight goal generation or curriculum-guided goal selection	achievability, usefulness, curriculum	changes goal selection	SABER keeps goal selection fixed
EBP	hindsight episode prioritization	trajectory energy	changes replay priority	SABER keeps uniform replay sampling
HGR	episode/goal ranking with IS correction	TD-error and sampling probability	changes sampling and critic gradient weight	SABER keeps replay sampling fixed and uses no IS ratio
MHER / MHER(λ)	multi-step hindsight target construction	n-step relabeled returns, λ-return	changes Bellman target	SABER keeps one-step HER target fixed
ARCHER	hindsight reward rescaling	real/hindsight reward identity	changes reward scale	SABER leaves rewards unchanged
USHER	HER bias-correcting Bellman reweighting	learned future-goal likelihood ratio	changes Bellman loss by IS ratio	SABER uses local action support without unbiasedness claim
PER / TD-error weighting	replay or loss prioritization	TD-error magnitude	changes sampling or loss by surprise	SABER weights by local replay-action support

2023). 이러한 방법은 SABER와 보완적이지만, update-to-data ratio와 critic ensemble architecture를 크게 바꾸므로 본 논문의 matched-backbone critic-weighting 분석과는 다른 compute/backbone regime에 속한다. 본 논문은 같은 HER backbone에서 updates per environment step만 늘린 high replay-ratio HER baselines를 포함해 update-scaling alternative를 확인하고, 추가로 FetchPickAndPlace에서 REDQ-HER target-Q-bound $N = 10, M = 2, \mathrm{UTD} = 4$ row를 직접 실행해 architecture/compute-scaled sparse-goal baseline을 보고한다.

SABER는 backbone 자체를 바꾸지 않는다. DDPG, TD3, SAC의 Bellman target 계산은 유지하되, critic regression loss에 sample-wise support-derived weight를 곱한다. 따라서 SABER는 새로운 actor-critic algorithm이라기보다, off-policy actor-critic에 부착되는 critic-side Bellman error reweighting layer로 보는 것이 정확하다.

Prioritized Replay and TD-Error Weighting

Prioritized Experience Replay (PER)는 TD-error가 큰 transition을 더 자주 sampling하여 learning progress가 클 것으로 예상되는 sample을 재사용한다 (Schaul et al. 2015). HGR도 TD-error를 hindsight goal ranking proxy로 사용한다 (Luu and Yoo 2021). PER-style loss weighting은 sampling을 바꾸지 않더라도 TD-error magnitude를 기준으로 critic loss를 조정한다.

SABER는 TD-error가 큰 sample을 우선시하지 않는다. SABER의 기준 신호는 local replay-action support이다. 즉, TD-error magnitude가 아니라 현재 policy action이 해당 state-goal-horizon context의 replay action support 안에 있는지를 본다. 또한 main SABER는 replay sampling distribution을 바꾸지 않고, critic loss weight만 바꾼다.

> SABER does not directly prioritize surprise; it uses local support as the weighting signal, which can still correlate with TD-error in practice.

Dataset Support and Extrapolation

Offline RL에서는 learned policy가 dataset support 밖의 action을 선택할 때 critic extrapolation error가 커질 수 있다는 문제가 잘 알려져 있다. BCQ와 CQL 등은 이러한 distribution shift와 OOD action overestimation을 줄이기 위해 policy constraint나 conservative value learning을 사용한다 (Fujimoto, Meger, and Precup 2019; Kumar et al. 2020).

SABER는 offline RL 방법이 아니다. SABER는 online off-policy HER setting에서 action을 제한하지 않으며, policy를 dataset support 안으로 강제로 묶지도 않는다. 대신 replay support가 충분한 Bellman update를 critic이 더 강하게 fitting하도록 weight를 조정한다. 따라서 SABER는 offline RL의 support mismatch 문제와 개념적으로 연결되지만, 알고리즘적 개입은 훨씬 좁다.

Broader Diagnostic Motivation

SABER는 broader diagnostic view of RL failures에서 동기를 얻었지만, 그 전체 framework를 구현한 것은 아니다. 여기서 필요한 부분은 agent가 충분히 관측했지만 policy/critic이 아직 효과적으로 활용하지 못하는 region을 더 안정적으로 학습시키는 practice-oriented intervention이다.

SABER는 이 아이디어를 sparse HER actor-critic setting으로 축소한다. World-model-based diagnosis나 full competence decomposition을 사용하지 않고, replay local support membership을 coarse proxy로 사용하며, focused data collection 대신 critic Bellman loss reweighting만 적용한다. 따라서 SABER는 diagnostic framework 전체가 아니라, HER 이후 critic fitting에 대한 좁은 instantiation이다.

Preliminaries

Goal-Conditioned MDP

Goal-conditioned MDP를 다음과 같이 둔다.

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{G}, P, r_g, \gamma). \tag{1}$$

Replay buffer에는 transition이 다음 형태로 저장된다.

$$\tau_i = (o_i, a_i^r, o_i', g_i^{env}, \tilde{d}_i, \chi_i, h_i), \tag{2}$$

여기서 o_i는 observation, a_i^r는 replay buffer에 저장된 executed action, o_i'는 next observation, g_i^{env}는 original environment goal, $\tilde{d}_i$는 environment가 반환한 raw done flag, χ_i는 time-limit timeout flag, h_i는 horizon 또는 time-to-goal context다. 아래에서는 critic Bellman regression이 replay action a_i^r에서 계산되고, support predicate는 current policy action a_i^π를 기준으로 계산된다는 점을 명시적으로 구분한다.

HER를 사용할 때 training goal은 original goal이 아니라 relabeling distribution에서 샘플링된다.

$$g_i^{tr} \sim \mathcal{H}(\tau_i). \tag{3}$$

HER를 사용하지 않는 경우에는 $g_i^{tr} = g_i^{env}$이다. 따라서 SABER는 HER 자체를 필수로 가정하기보다, training goal g_i^{tr}이 주어진 상태에서 작동하는 critic weighting method로 정의할 수 있다.

Goal-conditioned input을 다음처럼 둔다.

$$x_i = (o_i, g_i^{tr}). \tag{4}$$

Next-state input은 $x_i' = (o_i', g_i^{tr})$로 둔다. Bellman target 표기를 간단히 하기 위해

$$r_i = r(o_i, a_i^r, o_i', g_i^{tr}) \tag{5}$$

로 둔다. d_i는 raw done이 아니라 bootstrapping에 쓰는 terminal indicator를 뜻한다. 기본 Fetch/DrawerOpen 비교에서는 `success_terminates=false`, `timeout_bootstrap=true`이므로

$$d_i = \tilde{d}_i(1 - \chi_i) \tag{6}$$

이고, timeout transition은 bootstrapping을 허용하며 HER로 만들어진 success state도 terminal로 강제하지 않는다. Actor와 critic은 각각 다음과 같다.

$$\pi_\theta(x_i) = \pi_\theta(o_i, g_i^{tr}), \tag{7}$$

$$Q_\phi(x_i, a_i) = Q_\phi(o_i, g_i^{tr}, a_i). \tag{8}$$

Bellman Regression

DDPG-style deterministic actor-critic의 Bellman target은 다음과 같이 쓸 수 있다.

$$y_i = r_i + \gamma(1 - d_i) Q_{\bar{\phi}}(o_i', g_i^{tr}, \pi_{\bar{\theta}}(o_i', g_i^{tr})). \tag{9}$$

Baseline critic loss는 모든 sample을 동일 weight로 fitting한다.

$$\mathcal{L}_Q^{base}(\phi) = \frac{1}{B} \sum_{i=1}^{B} \left(Q_\phi(x_i, a_i^r) - y_i\right)^2. \tag{10}$$

TD3와 SAC에서는 target 계산 방식이 backbone에 맞게 달라지지만, critic이 Bellman target을 regression한다는 구조는 같다. TD3에서는 clipped target-policy smoothing과 twin critics를 사용한다.

$$\begin{aligned} \epsilon_i &\sim \mathcal{N}(0, \sigma^2), \\ \bar{\epsilon}_i &= \mathrm{clip}(\epsilon_i, -c, c), \\ \tilde{a}_i' &= \mathrm{clip}\left(\pi_{\bar{\theta}}(x_i') + \bar{\epsilon}_i, a_{\min}, a_{\max}\right), \end{aligned} \tag{11}$$

$$y_i^{TD3} = r_i + \gamma(1 - d_i) \min_{k \in \{1,2\}} Q_{\bar{\phi}_k}(x_i', \tilde{a}_i'). \tag{12}$$

SAC에서는 sampled next action과 entropy term이 target에 들어간다.

$$\tilde{a}_i' \sim \pi_\theta(\cdot \mid x_i'), \tag{13}$$

$$\begin{aligned} y_i^{SAC} = r_i + \gamma(1 - d_i)\big[&\min_{k \in \{1,2\}} Q_{\bar{\phi}_k}(x_i', \tilde{a}_i') \\ &- \alpha \log \pi_\theta(\tilde{a}_i' \mid x_i')\big]. \end{aligned} \tag{14}$$

SABER는 DDPG/TD3/SAC의 target construction을 바꾸지 않고, 아래에서 정의하는 sample-wise weight를 critic regression loss에만 곱한다.

Motivation: Target Quantity Is Not Critic-Fit Usefulness

HER는 sparse reward에서 target quantity를 늘린다. 하지만 target quantity와 current-policy action region에서의 critic-fit usefulness는 같은 개념이 아니다.

같은 mini-batch 안의 두 HER-relabeled transition을 생각하자. 두 transition 모두 reward를 재계산할 수 있고 Bellman target y_i를 만들 수 있다. 그러나 첫 번째 transition에서는 현재 actor action $\pi_\theta(o_i, g_i^{tr})$이 같은 local state-goal-horizon context 안에서 replay buffer에 여러 번 관측된 action family에 속할 수 있다. 두 번째 transition에서는 현재 actor action이 그 local context의 replay action support 바깥에 있을 수 있다.

두 경우 모두 Bellman target은 존재한다. 그러나 critic이 해당 target을 현재 policy improvement에 유용한 supervision으로 fitting할 때의 usefulness는 다를 수 있다. 첫 번째 경우는 local interpolation에 가깝고, 두 번째 경우는 extrapolation risk가 크다.

SABER의 기본 가설은 다음이다.

> Replay support가 충분한 transition은 current-policy action region과 더 직접적으로 연결된 critic fitting pressure를 제공한다. 따라서 support-positive transition의 critic TD loss를 더 크게 반영하면 sparse manipulation에서 value learning이 안정화되고 sample efficiency가 좋아질 수 있다.

이 가설은 theorem으로 주장하지 않는다. 본 논문에서는 이를 support-conditioned Bellman regression interpretation으로 제시하고, downstream performance, random controls, high replay-ratio baselines, and sample-level diagnostics로 평가한다. 현재 diagnostics는 ψ_i가 TD-error magnitude와 동일한 신호가 아님을 보이지만, support-positive samples가 본질적으로 lower-risk 또는 lower-variance Bellman targets라는 강한 결론까지는 제공하지 않는다. 더 정확한 해석은 support-conditioned rule이 critic-side Bellman error allocation을 바꾸고, 일부 backbone에서는 high-error updates에 대한 correction을 더 강하게 만든다는 것이다.

SABER

Overview

SABER는 각 training sample마다 local replay-action support를 계산하고, support-positive sample의 critic Bellman error에 더 큰 weight를 준다. Main variant는 critic-only이다. 절차는 다음과 같다.

1. HER 또는 baseline replay sampling으로 mini-batch를 가져온다.
2. 각 sample i에 대해 training goal g_i^{tr} 기준의 local context c_i를 만든다.
3. Replay buffer에서 같은 local context의 transition set $\mathcal{L}_i$를 찾는다.
4. Current policy action $a_i^\pi = \pi_\theta(o_i, g_i^{tr})$이 local replay action support 안에 있는지 확인한다.
5. Support-positive이면 $w_i = 1 + \lambda_S$, 아니면 $w_i = 1$을 부여한다.
6. Batch normalization으로 평균 weight를 1로 맞춘다.
7. Normalized weight를 critic Bellman error에만 곱한다.

Local Context

각 train item i에 대해 task-specific local context를 정의한다.

$$c_i = C(o_i, g_i^{tr}, h_i). \tag{15}$$

이 context는 observation, training goal, horizon information을 discrete hash로 변환한 것이다. 중요한 점은 replay buffer의 reference transition j도 자기 original goal이 아니라 query sample의 training goal g_i^{tr} 아래에서 다시 context 화된다는 것이다. 즉, local support는 "같은 relabeled goal 아래에서 비슷한 state-horizon region에 replay action이 있었는가"를 묻는다.

FetchPickAndPlace main experiments에서 사용한 exact context는 다음과 같다. Observation index 0:3은 end-effector position, 3:6은 object position, 9:11은 two-finger gripper state로 사용한다. Phase는 $d_{\text{hand,obj}} > d_{\text{reach}}$이면 approach, object height가 z_{lift} 이하이고 gripper gap이 threshold보다 크면 grasp_attempt, object height가 z_{lift} 이하이면 lift_needed, 나머지는 place_phase로 둔다. 이 phase 이름은 실제 contact state를 검증한 label이 아니라 intended control stage를 나타내는 discrete context label이다. Full context key는

$$\begin{aligned} C_{\text{full}}(o, g, h) =& 10^6 \,\text{phase} + 10^5 \,\text{bin}(d_{\text{hand,obj}}) \\ &+ 10^4 \,\text{bin}(d_{\text{obj,goal}}) + 10^3 \,\text{bin}(z_{\text{obj}}) \\ &+ 10^2 \,\text{bin}(\text{gripper_gap}) + \text{bin}(h). \end{aligned} \tag{16}$$

Bins는 hand-object distance $(0.03, 0.05, 0.08, 0.12, 0.20, 0.40)$, object-goal distance $(0.05, 0.10, 0.20, 0.40, 0.70)$, object height $(0.43, 0.48, 0.55, 0.65, 0.75)$, gripper gap $(0.02, 0.05, 0.08)$, horizon $(10, 25, 50, 100)$을 사용한다. Runtime optimization은 먼저 goal-independent coarse key

$$\begin{aligned} C_{\text{coarse}}(o, h) =& 10^4 \,\text{bin}(d_{\text{hand,obj}}) + 10^3 \,\text{bin}(z_{\text{obj}}) \\ &+ 10^2 \,\text{bin}(\text{gripper_gap}) + \text{bin}(h). \end{aligned} \tag{17}$$

로 후보를 제한한 뒤, C_{full}로 exact filtering을 수행한다.

Local Replay-Action Support

Sample i에 대한 local replay pool을 다음과 같이 둔다.

$$\mathcal{L}_i = \{j \in \mathcal{D} : C_{\text{full}}(o_j, g_i^{tr}, h_j) = C_{\text{full}}(o_i, g_i^{tr}, h_i)\}. \tag{18}$$

실제 구현에서는 C_{coarse} bucket에서 최대 N_{scan}개의 최근 후보를 deterministic-tail 방식으로 scan하고, 그 안에서 위의 exact full-context 조건을 만족하는 transition만 $\mathcal{L}_i$로 둔다.

같은 local context에서 관측된 replay action들을 action bin으로 변환한다.

$$\mathcal{B}_i = \{b(a_j^r) : j \in \mathcal{L}_i\}. \tag{19}$$

Continuous action은 각 dimension을 action range $[-1, 1]$로 normalize한 뒤, dimension별로 K개 bin으로 나누고 mixed-radix integer id로 합친다. FPP main experiments에서는 $K = 5$를 사용한다.

Current policy action과 그 bin은 다음과 같다.

$$a_i^\pi = \pi_\theta(o_i, g_i^{tr}), \tag{20}$$

$$b_i^\pi = b(a_i^\pi). \tag{21}$$

SABER는 b_i^π가 $\mathcal{B}_i$에 포함되는지를 통해 current policy action이 local replay support 안에 있는지 판정한다.

Policy-action support as a proxy. SABER weight는 current policy action a_i^π의 support를 보고 결정되지만, critic residual은 replay action a_i^r에서 계산된다. 따라서 SABER는 a_i^π의 support가 $Q_\phi(x_i, a_i^r)$ target noise를 직접 측정한다고 주장하지 않는다. 더 정확한 해석은 다음이다. Off-policy actor-critic에서 critic은 replay actions에서 Bellman regression을 수행하지만, actor update는 같은 x_i에서 current policy action의 value를 exploit한다. 그러므로 SABER의 support predicate는 "이 state-goal-horizon context가 current policy improvement에 필요한 action region을 replay buffer 안에서 어느 정도 갖고 있는가"를 나타내는 local-support proxy다. 이 proxy가 positive인 sample의 Bellman residual을 더 강하게 fitting하되, actor objective와 replay sampling은 바꾸지 않는다.

Support Indicator

Support-positive indicator는 다음과 같이 정의한다.

$$\begin{aligned} n_i^{avail} &= \min(|\mathcal{L}_i|, M_B), \\ \psi_i &= \mathbf{1}[|\mathcal{L}_i| \geq N_{\text{local}}] \\ &\quad \cdot \mathbf{1}[n_i^{avail} \geq M_{\text{avail}}] \cdot \mathbf{1}[b_i^\pi \in \mathcal{B}_i]. \end{aligned} \tag{22}$$

여기서 M_B는 local pool에서 support candidate로 사용할 수 있는 최대 개수이고, n_i^{avail}은 cap을 적용한 available local transition count다. M_{avail}은 그 available count의 최소 문턱이다. 구현 config의 legacy 이름은 `M_Q_min`이지만, 본 논문에서는 이것을 Q-value threshold나 unique action-bin diversity threshold로 해석하지 않는다. $\mathcal{B}_i$는 membership test를 위한 unique action-bin set이며, 현재 main SABER predicate는 $|\mathcal{B}_i|$에 대한 별도 diversity 조건을 두지 않는다.

따라서 main support predicate는 "exact local pool이 충분하고, cap 이후에도 최소 candidate count가 있으며, current policy action bin이 그 local replay-action bin set 안에 있는가"를 묻는다. FPP main experiments에서는 $M_B = 16$, $M_{\text{avail}} = 4$, $N_{\text{local}} = 4$, $N_{\text{scan}} = 512$, $\lambda_S = 1.0$, `weighting_mode=batch_norm`, `weight_scope=critic_only`을 사용한다. 이 값에서는 count 조건이 사실상 $|\mathcal{L}_i| \geq 4$로 축약되며, 나머지 결정적 조건은 $b_i^\pi \in \mathcal{B}_i$이다. 따라서 현재 구현의 support는 density나 action-bin diversity를 보장하는 강한 support estimate가 아니라 local action-bin membership support로 해석해야

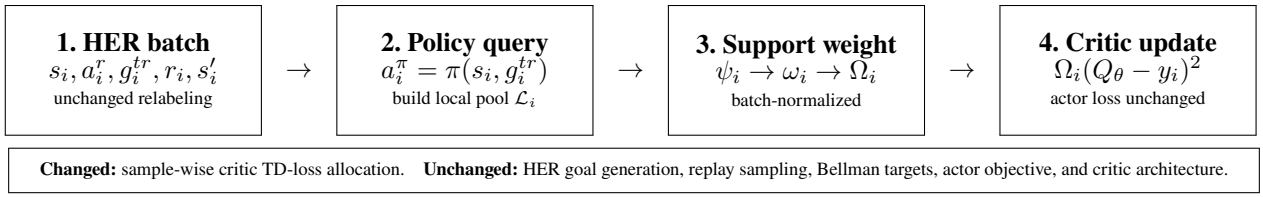


Figure 1: SABER intervenes after HER relabeling and before critic-loss aggregation. The current policy action is used only to query local replay-action support; Bellman regression remains at the replay action.

Table 2: Task-specific local context definitions used by SABER. Phase names denote intended control stages rather than verified physical contact states. M_{avail} corresponds to the legacy config name `M_Q_min`. All rows use action bins with $K = 5$ bins per action dimension and $\lambda_S = 1.0$.

Task	Full-context features	Phase predicate	Bin boundaries	Support settings
FetchPickAndPlace	phase; hand-object distance; object-goal distance; object z; gripper gap; horizon	approach if $d_{\text{hand,obj}} > 0.08$; else grasp_attempt if $z_{\text{obj}} \leq 0.48$ and gap > 0.04; else lift_needed if $z_{\text{obj}} \leq 0.48$; else place_phase	$d_{\text{hand,obj}}$: $(.03, .05, .08, .12, .20, .40)$; $d_{\text{obj,goal}}$: $(.05, .10, .20, .40, .70)$; z: $(.43, .48, .55, .65, .75)$; gap: $(.02, .05, .08)$; h: $(10, 25, 50, 100)$	$N_{\text{local}} = 4$, $M_{\text{avail}} = 4$, $M_B = 16$, $N_{\text{scan}} = 512$
FetchSlide	phase; ee-object distance; object-goal distance; object xy speed; velocity-goal alignment; horizon	near-goal if $d_{\text{obj,goal}} \leq 0.05$; else slide if speed ≥ 0.05; else strike if $d_{\text{ee,obj}} \leq 0.08$; else approach	$d_{\text{ee,obj}}$: $(.05, .10, .20, .40)$; $d_{\text{obj,goal}}$: $(.05, .10, .20, .40, .70, 1.00)$; speed: $(.02, .05, .10, .20, .40)$; alignment: $(-.50, 0, .50, .80)$; h: $(10, 25, 50, 100)$	$N_{\text{local}} = 4$, $M_{\text{avail}} = 4$, $M_B = 16$, $N_{\text{scan}} = 1024$ for DDPG row and 512 for TD3/SAC rows
Meta-World DrawerOpen	phase; hand-handle distance; handle-goal distance; handle y; gripper value; horizon	open if $d_{\text{handle,goal}} \leq 0.05$; else opening if $d_{\text{hand,handle}} \leq 0.08$ and $d_{\text{handle,goal}} \leq 0.12$; else contact if $d_{\text{hand,handle}} \leq 0.08$; else approach	$d_{\text{hand,handle}}$: $(.03, .05, .08, .12, .20, .40)$; $d_{\text{handle,goal}}$: $(.03, .05, .08, .12, .20, .40, .70)$; handle y: $(-.05, 0, .05, .10, .15, .25)$; gripper: $(-.50, 0, .50)$; h: $(10, 25, 50, 100, 200, 500)$	$N_{\text{local}} = 4$, $M_{\text{avail}} = 4$, $M_B = 16$, $N_{\text{scan}} = 512$

한다. 코드와 config에 남아 있는 N_{gap}은 older gap-weight variants의 gap-stat readiness threshold이며, main support-only/runtime-opt SABER predicate에는 직접 들어가지 않는다.

Weight Construction

Raw weight는 다음과 같다.

$$w_i^{raw} = 1 + \lambda_S \psi_i. \tag{23}$$

Batch-normalized weight는 다음과 같다.

$$\Omega_i = \frac{w_i^{raw}}{\frac{1}{B}\sum_{j=1}^{B} w_j^{raw} + \epsilon}. \tag{24}$$

Critic-only SABER에서는 다음과 같이 둔다.

$$\Omega_i^{critic} = \Omega_i, \qquad \Omega_i^{actor} = 1. \tag{25}$$

Critic Objective

SABER critic loss는 다음이다.

$$\mathcal{L}_Q^{SABER}(\phi) = \frac{1}{B}\sum_{i=1}^{B} \Omega_i \left(Q_\phi(x_i, a_i^r) - y_i\right)^2. \tag{26}$$

TD3/SAC처럼 twin critics를 사용하는 backbone에서는 같은 Ω_i를 두 critic residual에 모두 적용한다.

$$\mathcal{L}_Q^{SABER} = \frac{1}{B}\sum_i \Omega_i \Big[\left(Q_{\phi_1}(x_i, a_i^r) - y_i\right)^2 + \left(Q_{\phi_2}(x_i, a_i^r) - y_i\right)^2\Big]. \tag{27}$$

Actor objective는 backbone의 baseline actor objective를 유지한다. DDPG/TD3에서는 deterministic policy gradient 형태를 유지하고, SAC에서는 entropy-regularized actor objective를 유지한다. 구현상 SAC actor update는 Ω_i^{actor}를 사용하지 않는다.

TD-error and PER-style Loss Weighting Baselines

SABER가 TD-error magnitude 기반 weighting과 다른 신호를 사용하는지 확인하기 위해, uniform replay sampling을 유지한 PER-style loss-weighting baseline도 둔다. 먼저 replay action에서 TD-error magnitude를 계산한다.

$$\delta_i = \text{stopgrad}\left(|Q_\phi(x_i, a_i^r) - y_i|\right). \tag{28}$$

Twin-critic backbone에서는 두 critic residual의 minimum absolute error를 logging/priority용 δ_i로 사용한다. 이는

Algorithm 1 SABER Critic Update

Sample a mini-batch from replay buffer $\mathcal{D}$.
Assign training goals g_i^{tr} using HER or the original goals.
Compute backbone Bellman targets y_i.
for each sample i **do**
  Construct local context $c_i = C(o_i, g_i^{tr}, h_i)$.
  Retrieve local replay pool $\mathcal{L}_i$.
  Compute current policy action $a_i^\pi = \pi_\theta(o_i, g_i^{tr})$.
  Compute action-bin set $\mathcal{B}_i = \{b(a_j^r) : j \in \mathcal{L}_i\}$.
  Set $\psi_i = 1$ if $|\mathcal{L}_i| \geq N_{\text{local}}$, $\min(|\mathcal{L}_i|, M_B) \geq M_{\text{avail}}$, and $b(a_i^\pi) \in \mathcal{B}_i$; otherwise $\psi_i = 0$.
  Set $w_i^{raw} = 1 + \lambda_S \psi_i$.
end for
Normalize weights within the batch to obtain Ω_i.
Update the critic with weighted Bellman regression.
Update the actor using the unchanged baseline actor objective.

conservative priority proxy이며 canonical PER 구현의 유일한 선택은 아니다. Mean 또는 max twin residual을 쓰는 sensitivity는 아직 포함하지 않았으므로, 본 논문에서 PER-style row는 limited baseline으로 해석한다. 모든 TD-error-derived weight는 no-gradient로 계산하며, weight signal 자체에는 critic loss gradient를 흘리지 않는다. TD-error weighting baseline은 δ_i를 batch mean으로 normalize하고 clip한 뒤 $1 + \lambda_{TD}$ scale을 적용한다. PER-style loss weighting은 sampling distribution을 바꾸지 않고 다음 critic weight만 사용한다.

$$p_i = (\delta_i + \epsilon_{PER})^{\alpha_{PER}},$$
$$w_i^{PER} = \text{clip}\left(\frac{p_i}{\frac{1}{B}\sum_j p_j + \epsilon}, 0, U_{PER}\right). \tag{29}$$

그 뒤 w_i^{PER}도 SABER와 동일하게 batch-normalize하여 critic loss에만 곱한다. FPP PER-style baseline에서는 $\alpha_{PER} = 0.6$, $\epsilon_{PER} = 10^{-6}$, $U_{PER} = 2.0$, `weight_scope=critic_only`를 사용했다.

Theoretical Interpretation

SABER can be interpreted as support-conditioned Bellman regression with a deliberately narrow critic-fit allocation hypothesis. Let $\mathcal{U}_i^{fit}$ denote the policy-relevant usefulness of using transition i's replay-action Bellman residual as supervision for a critic that will later be queried by the actor at a_i^π. SABER uses the working hypothesis

$$\mathbb{E}[\mathcal{U}_i^{fit} \mid \psi_i = 1] > \mathbb{E}[\mathcal{U}_i^{fit} \mid \psi_i = 0]. \tag{30}$$

This is not a theorem in the current paper. It is an interpretation and empirical hypothesis. SABER does not claim that ψ_i directly measures Bellman target noise at a_i^r, and it does not claim that unsupported samples are useless or that support-positive samples have lower target variance. It only assigns stronger critic fitting weight to samples whose local context also supports the current policy action. The diagnostics therefore test whether ψ_i changes critic error allocation and residual correction, not whether it is a complete target-quality oracle.

This interpretation also explains why critic-only is the most isolating main variant. If support weights are applied to the actor loss, the method may change which state-goal regions the actor is encouraged to exploit, making it harder to separate critic fitting effects from actor distribution bias. Critic-only SABER is used for mechanism isolation: support affects Bellman regression, not direct policy optimization. This does not claim that critic-only is always the best-performing variant.

Experiments

Research Questions

본 실험은 다음 질문에 답하기 위해 설계한다.

- RQ1. SABER는 sparse-reward HER manipulation에서 baseline HER보다 mean sample efficiency와 mean final success를 높이는가?
- RQ2. SABER는 DDPG, TD3, SAC와 같은 여러 off-policy actor-critic backbone에 호환되는가?
- RQ3. SABER는 FetchPickAndPlace보다 다른 dynamics를 갖는 DrawerOpen에서도 큰 효과 크기와 seed consistency를 보이는가?
- RQ4. SABER의 이득은 현재 보유한 HGR-style sampler-only, PER-style, full-PER, high replay-ratio HER, REDQ-HER, shuffled, random-support, random-fixed-count controls로 설명될 수 있는가?
- RQ5. Sample-level diagnostics에서 support-positive indicator는 TD-error magnitude, high-error overlap, and residual correction과 어떤 관계를 갖는가?
- RQ6. SABER의 한계는 무엇이며, runtime overhead와 task/backbone-dependent failure는 어떻게 나타나는가?

Tasks and Metrics

주요 benchmark는 sparse-reward goal-conditioned manipulation tasks다. FetchPickAndPlace는 breadth evidence로 사용한다 (Plappert et al. 2018). 즉 DDPG, TD3, SAC 세 actor-critic backbone에서 같은 critic-side objective가 호환되는지 본다. FPP DDPG, TD3, SAC에서는 random-support 및 random-fixed-count controls를 추가해 random reweighting과 support-conditioned assignment의 차이를 비교한다. 또한 FPP DDPG에서는 high replay-ratio HER $4\times$와 $8\times$, FPP TD3와 SAC에서는 high replay-ratio HER $4\times$를 추가해 "critic update를 더 많이 하면 되는 것 아닌가"라는 compute-scaling alternative를 직접 확인한다. FPP에는 REDQ-HER target-Q-bound $N = 10, M = 2, \text{UTD} = 4$ row도 추가해 stronger architecture/compute-scaled sparse-goal reference를 둔다. Meta-World DrawerOpen DDPG는 effect-size evidence로 사용한다 (Yu et al. 2020). 이 task에서는 baseline DDPG+HER의 seed failure가 크고, SABER가 큰 mean improvement와 5/5 seed success를 보인다. 동일 DDPG, UTD 1, 2M-step budget에서 two-stage TD-error episode/goal ranking만 사용하는 HGR-style sampler-only control도 포함한다. 이 row는 canonical HGR의 episode/goal importance-sampling correction을 사용하지 않으므로 direct canonical-HGR comparison으로 해석하지 않는다. Random-Support,

PER-Style, Random-Fixed-Count도 강하므로 DrawerOpen DDPG는 HER 대비 large-effect evidence이지 structured support의 causal proof는 아니다. FetchSlide는 secondary evidence와 failure analysis로 사용한다. FetchPush artifact 도 별도로 보유하고 있지만 현재 review-only main draft 에는 complete and matched rows가 부족한 항목을 넣지 않는다. DrawerOpen TD3 actor/random controls와 DrawerClose DDPG/TD3 random/shuffled/PER-style controls는 보조 stress-test와 mechanism discussion으로 사용한다.

따라서 본 실험은 SOTA robotics benchmark competition이 아니라 matched-backbone critic-loss reweighting study로 설계한다. 핵심 비교는 같은 HER backbone, 같은 replay sampling, 같은 actor objective 아래에서 critic loss weight만 바꾸었을 때 sample efficiency와 control behavior가 어떻게 달라지는지다. High replay-ratio HER는 이 matched backbone 안에서 update-to-data ratio만 키우는 strong compute baseline으로 추가한다. REDQ-HER는 update-to-data ratio와 critic ensemble architecture까지 바꾸는 stronger architecture/compute-scaled reference로 포함한다. DrawerOpen sampler-only row는 sampling intervention stress control이며 canonical HGR baseline은 아니다; HGG와 MHER 같은 다른 goal-selection/target-construction alternatives도 Related Work에서 위치만 비교한다. 본 draft의 main claim은 이 방법들이나 REDQ-HER를 보편적으로 압도한다는 것이 아니다. 대신 SABER의 역할은 one-update matched HER baselines를 개선하는 critic-side allocation intervention을 분리하고, random-control 및 compute/backbone-scaling stress tests 아래에서 그 intervention의 효과와 한계를 보이는 것이다.

각 evaluation checkpoint의 success는 $E = 20$ evaluation episodes에서 계산한 environment success rate다. Seed s의 evaluation curve를 $\{(t_k, q_k^{(s)})\}_{k=1}^{K_s}$로 둔다. 여기서 t_k는 environment step이고 $q_k^{(s)} \in [0,1]$는 해당 checkpoint의 success rate다. Final success와 best success는 각각

$$\mathrm{Final}^{(s)} = q_{K_s}^{(s)}, \qquad \mathrm{Best}^{(s)} = \max_k q_k^{(s)} \tag{31}$$

로 계산한다.

Success AUC는 evaluation success curve의 normalized trapezoidal area다. 선택된 checkpoint index 집합 $I = \{j_1, \ldots, j_m\}$에 대해

$$\widehat{\mathrm{AUC}}^{(s)}(I; Z) = \frac{1}{Z} \sum_{\ell=1}^{m-1} \frac{q_{j_\ell}^{(s)} + q_{j_{\ell+1}}^{(s)}}{2} \left(t_{j_{\ell+1}} - t_{j_\ell}\right). \tag{32}$$

FPP DDPG/TD3와 DrawerOpen의 success AUC는 첫 번째와 마지막 selected evaluation checkpoint 사이의 span을 Z로 사용한다. FPP SAC는 legacy HER/SABER의 25k schedule과 newer controls의 10k schedule을 직접 섞지 않는다. 모든 SAC HER/SABER/random/high-RR/REDQ rows를 exact common grid $G = \{50\mathrm{k}, 100\mathrm{k}, \ldots, 1\mathrm{M}\}$의 20 checkpoints로 제한하고, interpolation, smoothing, forward filling, and extrapolation 없이 $Z = 950\mathrm{k}$로 AUC를 재계산한다. FPP SAC final success는 common grid의 1M checkpoint, Hit90은 이 grid에서 처음 0.90에 도달한 checkpoint를 사용한다. FetchSlide summary에는 같은 span-normalized AUC와 total-budget-normalized AUC가 모두 저장되어 있으므로, 본문 표는 task analyzer가 선택한 normalized AUC를 사용하고 cross-task AUC 절대값은 직접 비교하지 않는다. Last-quarter AUC는 $t_k \geq 0.75T$인 checkpoint만 사용하고 $Z = T/4$로 normalize한다.

Hit90은 threshold 0.90에 대한 first-hit metric이다.

$$H_{0.90}^{(s)} = \min\{t_k : q_k^{(s)} \geq 0.90\}. \tag{33}$$

도달하지 못한 seed의 $H_{0.90}^{(s)}$는 undefined로 둔다. FPP table은 reached seeds만으로 median first-hit step을 계산하고, 괄호에 reach count를 함께 보고한다. DrawerOpen DDPG table도 같은 방식으로 median first-hit step과 reach count를 함께 보고한다. 따라서 median first-hit step은 reach count와 함께 읽어야 하며, 서로 다른 reach count를 가진 methods 사이에서 censoring-free threshold metric으로 해석하면 안 된다. High-variance TD3 stress-test table은 compactness를 위해 Hit90 seed count만 보고한다.

모든 $\pm$ 값은 seeds 1–5에 대한 sample standard deviation이며 standard error가 아니다. 즉 metric $m^{(s)}$에 대해 $\bar{m} = \frac{1}{5}\sum_s m^{(s)}$와 $\sqrt{\frac{1}{4}\sum_s (m^{(s)} - \bar{m})^2}$를 보고한다. Steps/sec는 recorded wall-clock runtime으로 나눈 training environment steps이며 support lookup overhead를 포함한다. Old artifacts에서 evaluation/logging time 포함 여부가 runner마다 완전히 통일되어 있지는 않으므로, 이는 controlled wall-clock benchmark가 아니라 observed training throughput metric이다. Inference-time cost나 hardware-normalized wall-clock superiority claim으로 해석하지 않는다. Sample efficiency와 wall-clock efficiency는 분리해서 해석한다.

Reproducibility Snapshot

모든 main Fetch 실험은 uniform replay와 future-goal HER를 사용한다. HER ratio는 0.8이며, relabel된 sample의 reward는 $r(o_{t+1}, g_i^{tr})$로 다시 계산한다. Success state는 Bellman target에서 terminal로 강제하지 않고, timeout transition에는 bootstrapping을 허용한다. 즉, `success_terminates=false`, `timeout_bootstrap=true`를 기본 convention으로 둔다.

Actor와 critic은 hidden size 256의 two-layer MLP를 사용하고, optimizer는 Adam이다. 공통 hyperparameters는 $\gamma = 0.98$, $\tau = 0.005$, actor/critic learning rate 3×10^{-4}, batch size 256, replay buffer size 300k, warmup 5k steps, random start 10k steps, one update per environment step이다. High replay-ratio HER baselines만 예외적으로 updates per environment step을 4 또는 8로 늘리며, support lookup이나 SABER weight는 사용하지 않는다. REDQ-HER도 support lookup과 SABER weight를 사용하지 않지만, SAC-style actor에 10-critic ensemble, target subset $M = 2$, UTD 4, and sparse-reward target-Q bound $[-50, 0]$를 사용한다. DDPG와 TD3 exploration noise는 0.2다. TD3는 policy delay 2, target policy noise 0.2, target noise clip 0.5를 사용한다. SAC는 twin critics와 tanh Gaussian actor를 사용하며, initial $\alpha = 0.2$, automatic entropy tuning, target entropy $-|\mathcal{A}|$를 사용한다.

SABER main rows use $M_B = 16$, $M_{\mathrm{avail}} = 4$ (legacy config `M_Q_min=4`), $N_{\mathrm{local}} = 4$, $K = 5$ action bins per dimension, $\lambda_S = 1.0$, task/method-specific `local_search_max`, `support_scan_mode=deterministic_tail_v1`,

and batch-normalized critic-only weights. Replay sampling and actor objectives are unchanged from the corresponding HER backbone. Full reproducibility details are given in Appendix A.

Breadth Result: FetchPickAndPlace

FetchPickAndPlace는 backbone breadth를 보기 위한 main evidence다. 모든 결과는 1M environment steps, seeds 1–5 기준이다.

FPP에서는 SABER가 세 backbone 모두에서 mean final success와 mean success AUC를 높인다. DDPG에서는 final success mean이 0.940에서 0.960으로 소폭 상승하고 AUC mean이 0.539에서 0.609로 오른다. TD3에서는 final success mean이 0.840에서 0.970으로, AUC mean이 0.372에서 0.442로 오른다. Exact common-grid SAC에서는 final success mean이 0.790에서 0.990으로, AUC mean이 0.471에서 0.583으로 오른다. Paired bootstrap CI를 보면 TD3와 SAC의 AUC gain은 더 안정적이지만, DDPG의 SABER–HER AUC CI는 0을 포함한다. 따라서 FPP claim은 "all backbones show higher mean performance"로 해석하고, 모든 row에서 통계적으로 닫힌 universal gain으로 해석하지 않는다.

이 결과는 SABER가 하나의 특정 actor-critic implementation에만 붙는 trick이 아니라, 여러 off-policy actor-critic backbone에 호환되는 critic-side objective임을 보여준다. 다만 runtime은 감소하고, 아래 high replay-ratio baseline처럼 update-to-data ratio를 크게 키운 compute-heavy alternative와의 관계는 별도로 해석해야 한다. 따라서 FPP 결과는 matched one-update-per-step HER baselines 대비 environment-step sample efficiency evidence로 읽고, wall-clock cost, update budget, and confidence interval을 함께 보고해야 한다.

Strong Compute Baselines: High Replay-Ratio HER and REDQ-HER

SABER는 critic-side loss allocation을 바꾸지만 update-to-data ratio는 baseline과 동일하게 유지한다. 따라서 중요한 반문은 단순하다: 같은 HER backbone에서 critic/actor update를 더 많이 수행하면 SABER의 gain이 사라지는가? 이를 확인하기 위해 support lookup을 끄고, actor/critic architecture와 HER replay를 그대로 둔 채 updates per environment step만 늘린 high replay-ratio HER baselines를 실행했다. DDPG에서는 $4\times$와 $8\times$, TD3와 SAC에서는 $4\times$를 평가한다. 또한 stronger external sparse-goal baseline risk를 다루기 위해, SAC-style actor와 HER relabeling을 유지하되 10-critic REDQ ensemble, random target subset $M = 2$, UTD 4, target-Q bound를 사용하는 REDQ-HER row를 FetchPickAndPlace에서 추가한다. REDQ-HER는 SABER와 같은 matched-backbone regime이 아니라 architecture/compute-scaled reference다.

이 baseline은 SABER에 유리한 방향으로만 해석되지 않는다. High-RR $4\times$는 DDPG에서 SABER와 비슷한 environment-step AUC를 얻는다. 따라서 "DDPG SABER의 AUC gain은 어떤 compute scaling으로도 따라갈 수 없다"는 claim은 성립하지 않는다. 그러나 $4\times$는 약 3.98M gradient updates와 16.1 recorded training hours를 사용하며, SABER는 같은 1M updates와 약 6.7 hours에서 비슷한 AUC를 얻는다. 즉 DDPG에서 SABER의 위치는 high replay-ratio를 절대적으로 이기는 SOTA baseline이 아니라, far fewer updates로 comparable environment-step learning curve를 얻는 critic-side reweighting alternative다.

TD3에서도 비슷한 tradeoff가 나타난다. TD3 high-RR $4\times$는 AUC 0.449로 TD3 SABER의 AUC 0.442와 구분되지 않지만, final success는 0.610으로 낮고 Hit90은 1/5 seeds에서만 도달한다. Paired bootstrap에서 SABER–high-RR $4\times$ AUC delta는 -0.007 with CI $[-0.053, 0.027]$로 닫히지 않지만, final success delta는 $+0.360$ with CI $[0.140, 0.530]$이고 runtime은 약 2.0h 낮다. 따라서 high replay-ratio는 AUC를 높일 수 있는 strong compute baseline이지만, final success, seed consistency, update/runtime budget 측면에서는 SABER와 다른 tradeoff를 갖는다.

SAC에서는 compute scaling의 효과가 더 강하다. Exact common-grid에서 SAC high-RR $4\times$는 AUC 0.768로 SAC SABER의 AUC 0.583보다 높고, paired bootstrap의 SABER–high-RR $4\times$ AUC delta는 -0.184 with CI $[-0.235, -0.123]$로 음수다. 따라서 SAC에서 "SABER가 high replay-ratio HER보다 environment-step sample efficient하다"는 claim은 성립하지 않는다. 그러나 SAC high-RR $4\times$는 약 47.0 recorded training hours와 steps/sec 5.91을 사용하며, SABER는 약 9.6 hours와 steps/sec 29.1에서 final success 0.990을 얻는다. 이 row는 high replay-ratio가 strong compute-scaled baseline임을 보여주는 동시에, 본 논문의 scope가 compute scaling을 이기는 SOTA claim이 아니라 one-update-per-step matched-backbone critic allocation study임을 분명히 한다.

REDQ-HER target-Q-bound는 더 강한 external reference다. Exact common-grid에서 이 row는 final success 1.000, AUC 0.867, median Hit90 300k를 보이며, 5개 seed 모두 Hit100에도 도달한다. REDQ-HER–SABER AUC delta는 $+0.283$ with CI $[0.213, 0.329]$다. 따라서 FPP에서 REDQ-HER는 SABER와 high-RR SAC보다 강한 environment-step learning curve를 제공한다. 그러나 이 결과는 SABER의 matched-backbone claim을 반박하기보다 비교 regime을 분리한다. REDQ-HER는 10-critic ensemble, target subset $M = 2$, UTD 4, target-Q bound를 사용하고, 약 89.5 recorded training hours와 steps/sec 3.10을 기록한다. Target-Q clipping rate는 final summaries에서 거의 0에 가까워 target bound가 지속적으로 target을 누른 결과로만 보기는 어렵다. 즉 REDQ-HER는 strong architecture/compute-scaled baseline이며, SABER는 더 단순한 one-update matched-backbone critic allocation objective다. 본 논문은 SABER가 REDQ-HER를 이긴다고 주장하지 않는다.

DDPG high-RR $8\times$는 더 많은 updates가 항상 안정적인 개선으로 이어지지 않음을 보여준다. Mean AUC는 HER보다 낮고, 5개 seed 중 2개가 Hit90에 도달하지 못하며, 한 seed는 final success 0.0으로 collapse한다. 따라서 high update-to-data ratio는 SABER와 보완적인 strong baseline이지만, sparse HER setting에서 단순히 updates를 늘리는 것이 안정적인 해법이라고 보기는 어렵다.

FPP DDPG Controls: PER-Style, Shuffled, and Random Reweighting

PER-style weighting은 AUC에서 HER보다 높지만 SABER보다 낮고, final success mean은 HER보다도 낮다. Full PER는 high variance와 낮은 mean final success를 보인다. Direct TD-error loss-weighting run도 별도로 확인했지만, 현재 artifact는 모든 seed가 약 633k train steps에서 종료된

Table 3: FetchPickAndPlace main results at 1M environment steps. SAC AUC and Hit90 use the exact shared 50k–1M grid; DDPG/TD3 use their native common evaluation schedules. Hit90 reports the median first step among seeds that reached 0.90 success, with the reach count in parentheses. Steps/sec is observed training throughput, not a controlled wall-clock benchmark.

Backbone	Method	Final Success	Best Success	Success AUC	Median Hit90 Step	Steps/sec
DDPG	HER	0.940 ± 0.042	1.000 ± 0.000	0.539 ± 0.125	600k (5/5)	65.03 ± 0.10
DDPG	HER+SABER	0.960 ± 0.042	1.000 ± 0.000	0.609 ± 0.038	540k (5/5)	41.72 ± 0.20
TD3	HER	0.840 ± 0.188	0.880 ± 0.144	0.372 ± 0.032	980k (3/5)	65.33 ± 0.13
TD3	HER+SABER	0.970 ± 0.045	0.990 ± 0.022	0.442 ± 0.035	840k (5/5)	41.25 ± 0.59
SAC	HER	0.790 ± 0.288	0.910 ± 0.124	0.471 ± 0.079	800k (3/5)	38.93 ± 0.07
SAC	HER+SABER	0.990 ± 0.022	1.000 ± 0.000	0.583 ± 0.030	750k (5/5)	29.10 ± 0.12

Table 4: Paired bootstrap deltas for FPP SABER minus HER. Values report mean delta with 95% bootstrap confidence interval over seeds 1–5.

Backbone	Δ Final	Δ AUC
DDPG	$+0.020\,[-0.040, 0.070]$	$+0.070\,[-0.016, 0.193]$
TD3	$+0.130\,[-0.030, 0.310]$	$+0.071\,[0.032, 0.095]$
SAC	$+0.200\,[-0.020, 0.430]$	$+0.112\,[0.064, 0.161]$

partial run이므로 1M table에서는 제외한다. 따라서 현재 표는 SABER의 이득이 PER-style weighting이나 prioritized replay만으로 설명되기 어렵다는 제한적인 FPP-DDPG 근거로만 사용한다.

Random and shuffled controls make the mechanism interpretation more cautious. Random-fixed-count reaches the same final success mean as SABER but has AUC 0.540, almost identical to HER's 0.539. Paired bootstrap gives SABER–Random-fixed AUC delta $+0.069$ with 95% CI $[0.038, 0.103]$. Thus simply upweighting a fixed number of random samples per batch does not explain the early/mid training gain.

Random-support is much stronger. It preserves the number of support-positive samples per batch but randomly assigns the high weights, and reaches AUC 0.604, close to SABER's 0.609. The paired SABER–Random-support AUC delta is only $+0.005$ with CI $[-0.026, 0.036]$. Thus FPP DDPG does not isolate sample-level alignment as an independent mechanism. Shuffled-weight is intermediate: final success mean is SABER와 동일한 0.960이지만, AUC mean은 0.579로 SABER보다 낮고, median Hit90도 620k로 SABER의 540k보다 늦다. Paired bootstrap에서 SABER–Shuffled AUC delta는 $+0.030$이고 95% CI는 $[0.002, 0.064]$이다.

따라서 FPP controls가 닫아주는 결론은 좁다. SABER의 이득은 fixed-count random weighting만으로는 설명되지 않는다. 그러나 support-positive count 또는 support-conditioned reweighting regime 자체가 중요한 regularization으로 작동할 수 있으며, exact sample-level alignment의 독립 기여는 별도 sample-level diagnostic과 함께 해석해야 한다.

Internal FPP-DDPG sensitivity. Table 7 adds compact internal variants of the DDPG SABER row. These rows are not used as new primary baselines; they are included to show that the reported FPP result is not an artifact of hiding failed nearby variants. A simple lambda schedule is comparable to the main fixed-λ_S row, positive-MSE weighting is also close, while a positive-Huber variant and a 300k annealing variant are weaker. Together with the PER-style and full-PER controls in Table 6, these rows support a cautious sensitivity statement rather than a claim that the current hyperparameters are uniquely optimal.

다음 절에서는 이 해석을 보강하기 위해 instrumented FPP DDPG와 TD3 runs에서 support-positive indicator와 TD residual의 sample-level 관계를 직접 분석한다.

FPP TD3 Controls: Random Reweighting Does Not Explain the Gain

FPP DDPG controls만 보면 Random-support가 SABER와 거의 같은 AUC를 보여 exact sample-level alignment claim이 약해진다. 그러나 같은 FPP task에서 TD3 backbone으로 control을 반복하면 다른 양상이 나온다. Table 8에서 Random-support와 Random-fixed-count는 AUC와 Hit90 seed count 모두 HER와 비슷하거나 낮은 반면, SABER는 AUC와 final success를 모두 높인다.

Paired bootstrap confirms this separation. SABER–HER AUC delta is $+0.071$ with 95% CI $[0.032, 0.095]$. Random-support–HER AUC delta is only $+0.004$ with CI $[-0.042, 0.041]$, and Random-fixed-count–HER AUC delta is -0.002 with CI $[-0.034, 0.024]$. Directly comparing against controls, SABER–Random-support AUC delta is $+0.067$ with CI $[0.052, 0.078]$, and SABER–Random-fixed-count AUC delta is $+0.073$ with CI $[0.054, 0.091]$. Thus, for FPP TD3, support-conditioned SABER is not interchangeable with fixed-count random reweighting or random assignment of support-count weights.

FPP SAC Controls: Final Success Saturation but AUC Separation

SAC adds a third backbone-level control check. In this setting Random-support remains close to HER in AUC, while Random-fixed-count reaches saturated final success but still

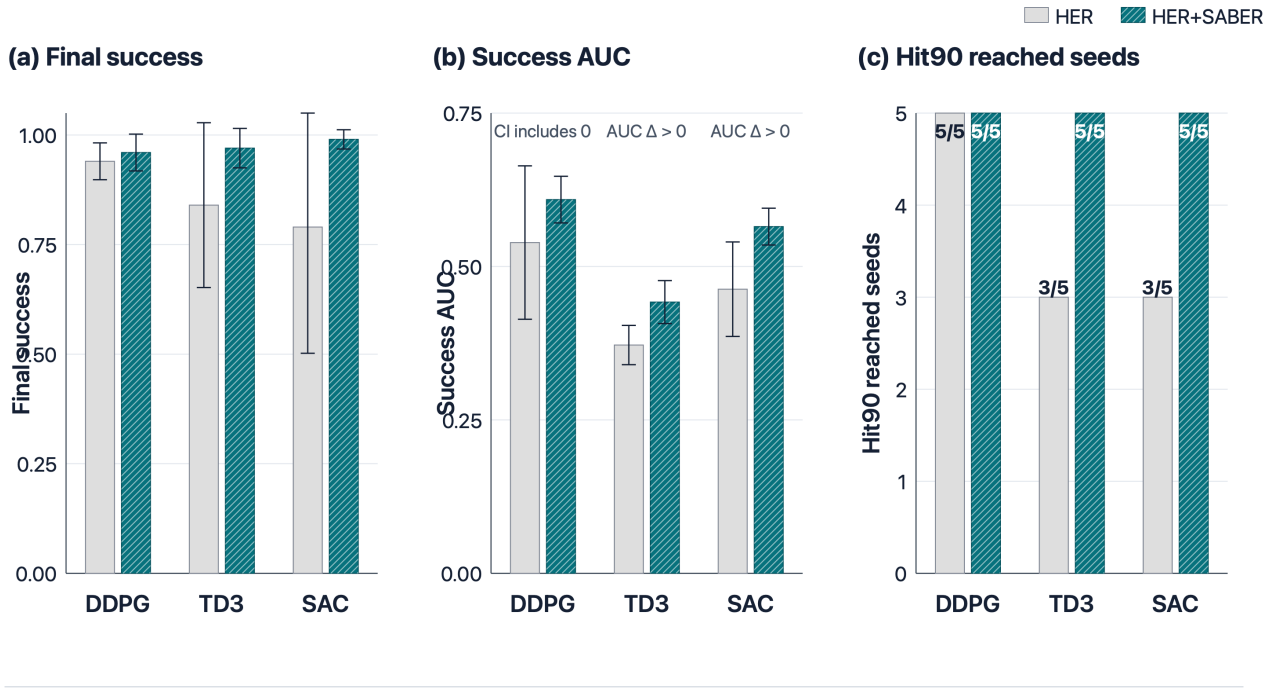


Figure 2: FetchPickAndPlace evidence across matched one-update-per-step HER backbones. Bars show the seed mean and standard deviation. SABER has higher means on final success and AUC for DDPG, TD3, and SAC; the paired AUC interval includes zero for DDPG and excludes zero for TD3/SAC. Scaling references are reported separately.

lags SABER in AUC. Thus SAC does not support a simple "random reweighting is always enough" explanation, but it also cautions against using final success alone as the mechanism metric.

Paired bootstrap shows the same split. SABER–Random-support has final delta $+0.300$ with CI $[0.110, 0.490]$ and common-grid AUC delta $+0.101$ with CI $[0.034, 0.164]$. Against Random-fixed-count, SABER has slightly lower final success delta -0.010 with CI $[-0.030, 0.000]$, but higher common-grid AUC delta $+0.058$ with CI $[0.003, 0.113]$. Therefore SAC provides additional evidence that count-matched random weighting can be a strong regularizer, especially for final success, but it does not fully account for SABER's early/mid-training AUC gain.

The SAC high replay-ratio and REDQ-HER rows change a different claim. They are not random-control rows; they are compute/backbone-scaled HER baselines. SAC high-RR $4\times$ reaches higher AUC than SABER, and REDQ-HER TQB is stronger still in FPP AUC. These rows should not be used to claim SABER dominance over update or architecture scaling. Instead, the random-control rows support the critic-reweighting mechanism boundary, while the scaling rows show that substantially larger update/backbone budgets can buy better environment-step AUC at much larger observed wall-clock cost.

Taken together, FPP DDPG, TD3, and SAC show task/backbone-dependent mechanism evidence. DDPG suggests that support-conditioned reweighting regime or count-preserving randomization can act as a strong regularizer. TD3 shows a case where the structured support-conditioned assignment separates clearly from random controls. SAC is intermediate for random controls: fixed-count random weighting saturates final success, but SABER remains higher in AUC. At the same time, SAC high-RR $4\times$ shows that update scaling can outperform SABER in environment-step AUC. Therefore the current evidence supports a cautious mechanism statement: random reweighting contributes in some settings, update scaling can be very strong, but neither random-control evidence nor high-RR evidence reduces SABER to a trivial fixed-count reweighting rule under the standard one-update-per-step matched-backbone setting.

FPP Sample-Level Diagnostics: DDPG, TD3, and SAC

FPP random-control 결과만으로는 support-conditioned weighting과 TD-error/PER-style weighting의 관계를 충분히 닫기 어렵다. 이를 확인하기 위해 FPP DDPG, TD3, and SAC SABER를 1M steps, seeds 1–5로 다시 실행하면서 10k steps마다 현재 training batch의 sample-level diagnostic을 기록했다. 각 checkpoint에서는 full batch 256개 sample에 대해 ψ_i, update 전후 $|\delta_i|$, top-k TD-error overlap, target variance를 계산했고, 별도로 128개 sample row를 저장했다. 각 backbone에서 500 diagnostic batches와 64k logged sample rows를 얻었다. SAC diagnostic run is used for mech-

Table 5: FPP high replay-ratio and REDQ-HER baselines at 1M environment steps. High-RR rows keep the same backbone and replay sampling fixed but increase updates per environment step. REDQ-HER additionally changes the critic architecture to a 10-critic ensemble with target subset $M = 2$ and target-Q bounding. SAC/REDQ AUC and Hit90 use the exact shared 50k–1M grid. Runtime is observed training wall-clock time, not a hardware-normalized benchmark.

Backbone	Method	Updates/step	Final	AUC	Hit90	Runtime h
DDPG	HER	1	0.940 ± 0.042	0.539 ± 0.125	600k (5/5)	4.27
DDPG	HER+SABER	1	0.960 ± 0.042	0.609 ± 0.038	540k (5/5)	6.66
DDPG	HER high-RR $4\times$	4	0.990 ± 0.022	0.618 ± 0.071	480k (5/5)	16.11
DDPG	HER high-RR $8\times$	8	0.720 ± 0.407	0.443 ± 0.269	620k (3/5)	31.85
TD3	HER	1	0.840 ± 0.188	0.372 ± 0.032	980k (3/5)	4.25
TD3	HER+SABER	1	0.970 ± 0.045	0.442 ± 0.035	840k (5/5)	6.74
TD3	HER high-RR $4\times$	4	0.610 ± 0.243	0.449 ± 0.079	630k (1/5)	8.76
SAC	HER	1	0.790 ± 0.288	0.471 ± 0.079	800k (3/5)	7.14
SAC	HER+SABER	1	0.990 ± 0.022	0.583 ± 0.030	750k (5/5)	9.55
SAC	HER high-RR $4\times$	4	0.960 ± 0.042	0.768 ± 0.097	450k (5/5)	47.02
REDQ-SAC	REDQ-HER TQB	4	1.000 ± 0.000	0.867 ± 0.052	300k (5/5)	89.52

Table 6: FPP DDPG controls at 1M steps. Random-support preserves the number of support-positive samples per batch but randomly assigns high weights. Random-fixed-count assigns high weights to $k = 94$ random samples per batch, matching the approximate mean support-positive count but using no support lookup. Random-fixed-count rules out a pure fixed-count explanation for the DDPG AUC gain, while Random-support prevents attributing that gain to exact sample-level alignment alone.

Method	Final	AUC	Median Hit90 Step	Steps/sec
DDPG+HER	0.940 ± 0.042	0.539 ± 0.125	600k (5/5)	65.03 ± 0.10
DDPG+HER+SABER	0.960 ± 0.042	0.609 ± 0.038	540k (5/5)	41.72 ± 0.20
DDPG+HER+Random-support	0.950 ± 0.050	0.604 ± 0.016	540k (5/5)	41.63 ± 0.26
DDPG+HER+Shuffled-weight	0.960 ± 0.055	0.579 ± 0.034	620k (5/5)	41.58 ± 0.10
DDPG+HER+Random-fixed-count	0.960 ± 0.042	0.540 ± 0.017	630k (5/5)	64.52 ± 0.05
DDPG+HER+PER-style	0.930 ± 0.104	0.568 ± 0.069	630k (5/5)	61.07 ± 0.09
DDPG+HER+PER-full	0.780 ± 0.438	0.460 ± 0.238	640k (4/5)	58.02 ± 0.14

anism analysis only; on the exact 50k common grid it reaches final success 0.990 ± 0.022, AUC 0.583 ± 0.030, and median Hit90 750k (5/5), matching the main SAC SABER curve on that grid.

This diagnostic gives a more precise claim boundary. The support predicate is not orthogonal to TD-error: support-positive samples are mildly enriched among high-TD-error samples, and they have larger average $|\delta|$ before the update. However, the relation is weak rather than equivalent: Pearson and Spearman correlations are only about 0.12, and top-10% overlap is 0.516 rather than close to 1.0. Thus SABER cannot be reduced to selecting the largest TD-error samples, but the FPP DDPG signal also cannot be described as independent of TD-error.

The residual-reduction results are also mixed. Support-positive samples show larger raw $|\delta|$ reduction, but the per-weight reduction $\Delta|\delta|/\Omega$ is close between groups. Therefore this diagnostic supports the critic-side reweighting interpretation but does not prove that support-positive samples are intrinsically lower-risk Bellman targets. Combined with the random-control results, the most defensible conclusion is that FPP DDPG SABER uses a weakly TD-error-correlated support-conditioned signal that changes critic fitting pressure; it is not merely PER-style surprise prioritization, but it is also not a pure lower-noise proof.

TD3 provides a cleaner diagnostic target because the FPP TD3 random controls remain near HER while SABER improves AUC. We therefore repeated the same instrumentation for TD3+HER+SABER. The diagnostic run exactly matches the original TD3 SABER aggregate performance within the stored summaries: final success 0.970 ± 0.045, AUC 0.442 ± 0.035, and Hit90 in 5/5 seeds. Thus the instrumentation is observational and does not change the learning result.

The TD3 diagnostic strengthens the claim that SABER is not merely random reweighting. Support-positive samples make up about 22% of the batch, but 36% of the top-10% TD-error samples are support-positive, corresponding to 1.67x enrichment. The signal is still not identical to TD-error: Pearson correlation is only 0.128 and top-error overlap is far from 1.0. However, unlike the DDPG diagnostic, TD3 shows both larger raw residual reduction and larger $\Delta|\delta|/\Omega$ for support-positive samples. This suggests that in the TD3 setting where random controls fail to match SABER, the

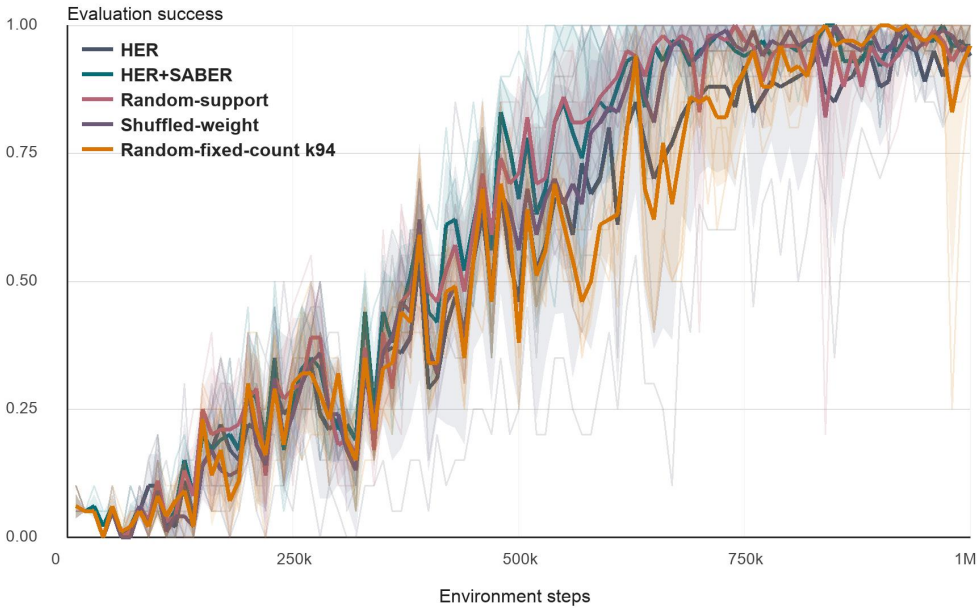


Figure 3: FetchPickAndPlace DDPG learning curves for HER, HER+SABER, Random-support, Random-fixed-count, and Shuffled-weight controls. Thin lines show individual seeds, and shaded regions show across-seed standard deviation. Random-support tracks SABER closely, so this figure supports a cautious support-conditioned reweighting interpretation rather than a pure sample-level alignment proof.

Table 7: FPP DDPG internal sensitivity rows at 1M steps. All rows use DDPG+HER with critic-only weighting and seeds 1–5. The table is appendix-style evidence for implementation sensitivity; it is not used to claim universal optimality of any single weighting variant.

Variant	Final	AUC	Median Hit90 Step	Steps/sec
Main SABER, fixed $\lambda_S = 1.0$	0.960 ± 0.042	0.609 ± 0.038	540k (5/5)	41.72 ± 0.20
Lambda schedule	0.950 ± 0.087	0.611 ± 0.038	530k (5/5)	41.65 ± 0.18
Quality floor 0.25	0.980 ± 0.045	0.581 ± 0.041	590k (5/5)	38.82 ± 0.13
Anneal 300k	0.930 ± 0.104	0.532 ± 0.134	590k (5/5)	41.53 ± 0.80
Positive-Huber weighting	0.770 ± 0.217	0.327 ± 0.070	885k (4/5)	39.11 ± 0.37
Positive-MSE weighting	0.990 ± 0.022	0.593 ± 0.052	550k (5/5)	40.37 ± 0.37

support-conditioned assignment identifies a subset that is both high-error enriched and more strongly corrected by the critic update.

At the same time, TD3 target variance is higher for support-positive samples. Thus the diagnostic still does not justify a lower-noise target claim. SAC provides a third-backbone diagnostic with the same broad pattern.

SAC strengthens the cross-backbone diagnostic story without changing the claim boundary. Like TD3, SAC shows top-error enrichment and larger immediate residual correction for support-positive samples, including a larger $\Delta|\delta|/\Omega$. Like DDPG and TD3, the signal is still not equivalent to TD-error ranking: Pearson correlation is only 0.143, and the top-10% overlap is far from 1.0. Also like the other diagnostics, support-positive target variance is higher, not lower. The safer mechanism statement is therefore: SABER changes critic-side Bellman error allocation toward a support-conditioned subset that is weakly correlated with TD-error, enriched among high-error samples, and in TD3/SAC receives larger immediate residual correction. This is a useful mechanism diagnostic, not a theorem that support-positive samples are intrinsically lower-noise.

Additional rows reinforce the same cautious boundary. DDPG high-RR $8\times$ drops to AUC 0.443 with 3/5 Hit90 seeds and one collapsed seed, so increasing updates is not monotonically stabilizing. DrawerClose DDPG and TD3 are treated as high-variance stress tests rather than primary statistically closed improvement claims; in these stress tests, random/fixed/shuffled controls remain competitive with SABER in at least one metric. FetchSlide DDPG degrades in final success and late AUC, so SABER is not a universal improvement over HER.

Table 8: FPP TD3 random controls at 1M steps. Random-support preserves the number of support-positive samples per batch but randomly assigns high weights. Random-fixed-count assigns high weights to $k = 73$ random samples per batch, matching the approximate mean TD3 SABER support-positive count but using no support lookup. Here both random controls remain near HER, ruling out a random-reweighting-only explanation for the TD3 gain.

Method	Final	AUC	Median Hit90 Step	Steps/sec
TD3+HER	0.840 ± 0.188	0.372 ± 0.032	980k (3/5)	65.32 ± 0.13
TD3+HER+SABER	0.970 ± 0.045	0.442 ± 0.035	840k (5/5)	41.25 ± 0.59
TD3+HER+Random-support	0.690 ± 0.284	0.375 ± 0.051	925k (2/5)	40.81 ± 0.32
TD3+HER+Random-fixed-count	0.670 ± 0.228	0.370 ± 0.032	775k (2/5)	65.21 ± 0.08

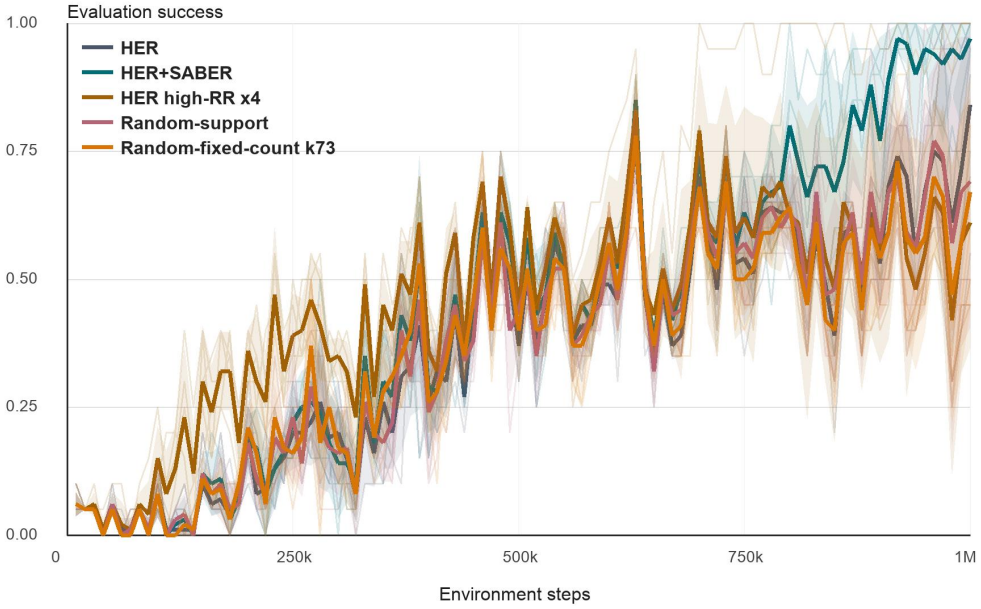


Figure 4: FetchPickAndPlace TD3 learning curves for HER, HER+SABER, Random-support, and Random-fixed-count controls. In contrast to FPP DDPG, the random controls remain close to HER while SABER improves AUC and reaches Hit90 in all five seeds.

Large-Effect Result: Meta-World DrawerOpen DDPG

DrawerOpen DDPG는 FPP breadth result와 다른 역할을 한다. FPP가 여러 backbone에 걸친 compatibility를 보여준다면, DrawerOpen DDPG는 sparse manipulation setting에서 critic-side reweighting이 HER 대비 큰 효과 크기와 seed consistency를 만들 수 있음을 보여준다. 함께 보고하는 HGR-style sampler-only control은 episode mean TD-error와 candidate future-goal TD-error를 이용해 HER sampling을 바꾸지만 critic loss weight는 모두 1로 유지한다. Canonical HGR의 importance-sampling correction을 생략하므로 이 row를 direct HGR baseline으로 부르지 않는다. Random-Support, PER-Style, Random-Fixed-Count도 함께 비교하므로, 이 절은 structured support의 causal proof가 아니라 critic-side allocation의 effect size와 seed robustness를 sampling and reweighting alternatives 아래에서 평가하는 결과로 읽어야 한다. 모든 결과는 2M environment steps, seeds 1–5 기준이다.

SABER improves DrawerOpen DDPG by a large margin: final success mean increases from 0.390 to 1.000, and AUC increases from 0.207 to 0.754. Paired bootstrap gives SABER–HER final delta $+0.610$ with 95% CI $[0.210, 1.000]$, and AUC delta $+0.546$ with CI $[0.340, 0.741]$. Hit90 also changes from 2/5 reached seeds to 5/5 reached seeds.

The HGR-style sampler-only row is a TD-error sampling ablation, not a faithful canonical HGR baseline. It raises mean AUC from HER's 0.207 to 0.311 and, on its two successful seeds, reaches Hit90 at 430k and 470k steps. However, the other three seeds remain at zero through 2M steps, giving final success 0.400 ± 0.548, AUC 0.311 ± 0.425, and only 2/5 Hit90 coverage. Its paired AUC delta over HER is $+0.103$ with CI $[-0.281, 0.462]$. SABER minus this sampler-only row gives final-success delta $+0.600$ with CI $[0.200, 1.000]$ and AUC delta $+0.443$ with CI $[0.106, 0.780]$. These numbers characterize the current ablation only and do not support superiority over canonical HGR.

The added controls make the mechanism interpretation stricter. Random-support reaches the same final success as SABER and AUC 0.706; SABER–Random-support AUC delta is $+0.047$ with CI $[-0.042, 0.136]$. PER-style

Table 9: FPP SAC random controls at 1M steps on the exact shared 50k–1M evaluation grid. Random-support preserves the support-positive count per batch but randomly assigns high weights. Random-fixed-count assigns high weights to $k = 76$ random samples per batch, matching the approximate mean SAC SABER support-positive count but using no support lookup. Random-fixed-count saturates final success, but SABER remains higher in AUC.

Method	Final	AUC	Median Hit90 Step	Steps/sec
SAC+HER	0.790 ± 0.288	0.471 ± 0.079	800k (3/5)	38.93 ± 0.07
SAC+HER+SABER	0.990 ± 0.022	0.583 ± 0.030	750k (5/5)	29.09 ± 0.12
SAC+HER+Random-support	0.690 ± 0.263	0.482 ± 0.069	725k (2/5)	28.86 ± 0.28
SAC+HER+Random-fixed-count	1.000 ± 0.000	0.526 ± 0.059	900k (5/5)	38.67 ± 0.03

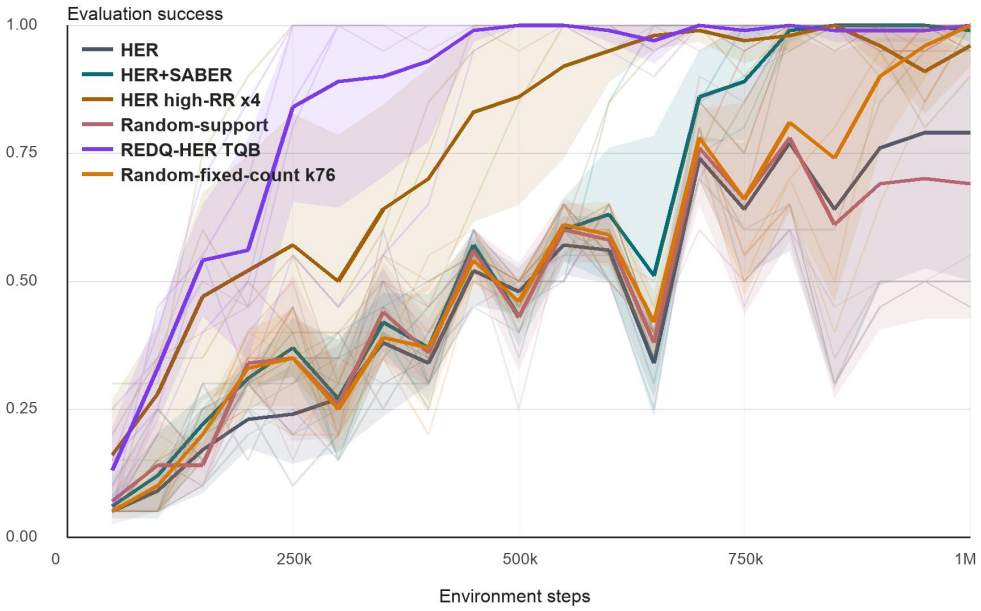


Figure 5: FetchPickAndPlace SAC learning curves on the exact shared 50k–1M grid. The plot includes HER, HER+SABER, Random-support, Random-fixed-count, high-RR $4\times$, and REDQ-HER TQB. Scaling references learn faster in environment steps; among one-update random controls, SABER has the highest AUC.

also improves strongly over HER, reaching final success 0.800 and AUC 0.715; SABER–PER-style AUC delta is $+0.038$ with CI $[-0.015, 0.091]$. Random-fixed-count also improves over HER, reaching final success 0.800 and AUC 0.621, but remains less consistent than SABER: one seed fails completely and the AUC standard deviation is 0.352. Among reached seeds, the sampler-only row and Random-fixed-count reach Hit90 earlier than SABER, so SABER should not be described as uniformly faster in first-hit time. Shuffled-weight reaches final success 0.400 and AUC 0.400, with only 2/5 seeds reaching Hit90. Therefore DrawerOpen DDPG supports the practical value and seed consistency of critic-side reweighting over HER, while showing that sampling, random/support-count assignment, and TD-error-style reweighting can contribute. The defensible claim is a seed-consistent large-effect result in this task, not superiority over canonical HGR or every alternative.

FetchSlide: TD3/SAC Transfer and DDPG Failure

FetchSlide는 SABER의 장점과 한계를 동시에 보여주는 중요한 secondary result다. 모든 결과는 1M steps, seeds 1–5 기준이다.

FetchSlide에서 TD3+SABER는 TD3+HER보다 mean final success, best success, success AUC, last-quarter AUC가 모두 높다. SAC+SABER도 SAC+HER보다 mean final success와 AUC가 소폭 높다. 그러나 DDPG+SABER는 DDPG+HER보다 final success, success AUC, last-quarter AUC가 모두 낮다.

이 결과는 SABER가 모든 task/backbone 조합에서 항상 개선된다는 주장을 피해야 함을 보여준다. 또한 FetchSlide artifacts는 DDPG row에서 $N_{\text{scan}} = 1024$, TD3/SAC rows에서 $N_{\text{scan}} = 512$를 사용하므로 strictly matched support-budget comparison으로 해석하지 않는다. 더 정확한 해석은 다음이다: SABER is compatible with multiple off-policy actor-critic backbones and shows higher mean performance on FetchPickAndPlace, while FetchSlide reveals task/backbone-dependent limitations.

Table 10: FPP DDPG SABER sample-level diagnostic over 1M steps and seeds 1–5. Values report the mean and across-seed standard deviation of per-seed diagnostic means. Correlations use full diagnostic batches, not only the logged 128-sample subset.

Metric	Mean ± seed std	Interpretation
Support-positive rate	0.367 ± 0.011	about 37% of batch samples receive high raw weight
$\mathrm{corr}(\psi_i, \lvert\delta_i\rvert)$, Pearson	0.116 ± 0.016	weak positive relation with TD-error magnitude
$\mathrm{corr}(\psi_i, \lvert\delta_i\rvert)$, Spearman	0.125 ± 0.017	weak rank relation
Top-10% TD-error support overlap	0.516 ± 0.022	enriched over the 0.367 support base rate
Top-20% TD-error support overlap	0.487 ± 0.016	enriched but not identical to high-TD-error selection
$\lvert\delta\rvert_{\mathrm{before}}$, support-positive	0.686 ± 0.044	support-positive samples tend to have larger residuals
$\lvert\delta\rvert_{\mathrm{before}}$, support-negative	0.484 ± 0.052	lower mean residual than support-positive group
$\Delta\lvert\delta\rvert$, support-positive	0.019 ± 0.003	larger raw residual reduction after weighted update
$\Delta\lvert\delta\rvert$, support-negative	0.008 ± 0.002	smaller raw residual reduction
$\Delta\lvert\delta\rvert/\Omega$, support-positive	0.013 ± 0.002	per-weight reduction close to unsupported group
$\Delta\lvert\delta\rvert/\Omega$, support-negative	0.011 ± 0.003	suggests weight scale contributes to raw reduction
Target variance, support-positive	230.4 ± 28.2	support-positive targets are not lower-variance targets
Target variance, support-negative	115.6 ± 16.2	lower target variance in this diagnostic

Table 11: FPP TD3 SABER sample-level diagnostic over 1M steps and seeds 1–5. TD3 uses twin critics; $|\delta|$ reports the mean absolute residual across the two critics. Values are means and across-seed standard deviations of per-seed diagnostic means.

Metric	Mean ± seed std	Interpretation
Support-positive rate	0.219 ± 0.008	about 22% of batch samples receive high raw weight
$\mathrm{corr}(\psi_i, \lvert\delta_i\rvert)$, Pearson	0.128 ± 0.012	weak positive relation with TD-error magnitude
$\mathrm{corr}(\psi_i, \lvert\delta_i\rvert)$, Spearman	0.179 ± 0.019	weak-to-moderate rank relation
Top-10% TD-error support overlap	0.360 ± 0.014	enriched over the 0.219 support base rate
Top-10% TD-error enrichment	1.67 ± 0.07	support-positive samples appear 1.67x more often in top-error set
Top-20% TD-error support overlap	0.341 ± 0.015	enriched but not equivalent to TD-error selection
$\lvert\delta\rvert_{\mathrm{before}}$, support-positive	0.741 ± 0.039	larger pre-update residual than support-negative group
$\lvert\delta\rvert_{\mathrm{before}}$, support-negative	0.456 ± 0.031	lower mean residual
$\Delta\lvert\delta\rvert$, support-positive	0.021 ± 0.001	larger raw residual reduction after weighted update
$\Delta\lvert\delta\rvert$, support-negative	0.006 ± 0.003	smaller raw residual reduction
$\Delta\lvert\delta\rvert/\Omega$, support-positive	0.013 ± 0.001	larger correction remains after weight normalization
$\Delta\lvert\delta\rvert/\Omega$, support-negative	0.008 ± 0.004	lower per-weight correction
Target variance, support-positive	418.6 ± 22.9	support-positive targets are not lower-variance targets
Target variance, support-negative	227.4 ± 14.3	lower target variance in this diagnostic

Additional DrawerOpen TD3 Stress-Test and Controls

DrawerOpen TD3는 DDPG large-effect result를 보조하는 stress-test로 사용한다. 여기서는 TD3+HER와 TD3+HER+SABER를 2M steps, seeds 1–5로 비교한다. DDPG result와 달리 variance가 매우 크므로, 이 section은 main proof가 아니라 limitation과 control 해석에 가깝다.

DrawerOpen TD3에서 SABER는 TD3+HER보다 mean final success, best success, AUC, Hit90 seed count가 높다. 그러나 variance가 매우 크고 paired uncertainty도 크다. 따라서 이 TD3 row는 DDPG DrawerOpen의 large-effect evidence를 보조하는 외부 stress-test로 해석한다.

Actor-Scope and Random Controls

Actor+Critic은 aggregate AUC에서 critic-only보다 높지만, high variance 아래에서 clean dominance를 보이지 않는다. Actor-only는 더 약하다. Critic-only는 actor update를 바꾸지 않기 때문에, support-conditioned weight가 Bellman regression에 미치는 효과를 가장 깨끗하게 분리한다.

TD3 controls도 mechanism claim을 조심스럽게 만든다. Random-Support가 Critic-Only와 비슷하거나 더 높은 AUC를 보였다는 점은 exact structured support rule만이 유일한 원인이라는 강한 claim을 약화시킨다. DrawerOpen TD3에서는 Shuffled-Weight가 실패하지만, FPP에서는 Shuffled-Weight control이 strong final success를 보이고 DrawerOpen DDPG에서는 Random-fixed-count도 HER보다 높다. 따라서 random/shuffled/fixed-count control의 의미는 task-dependent이며, 현재 결과만으로 structured support를 유일한 mechanism으로 분리할 수 없다. 이 결과들은 mechanism proof가 아니라 limitation/discussion evidence로 해석한다.

Additional DrawerClose TD3 Stability Stress-Test

DrawerClose TD3는 DrawerOpen과 반대 방향의 manipulation dynamics를 갖는 high-variance stress-test다. 이 결과는 main improvement claim이 아니라 stability와 random-control 해석을 보조하기 위해 appendix-style로 제시한다. 모든 methods는 2M steps, seeds 1–5로 실행했다.

DrawerClose TD3의 핵심은 early success가 아니라 final stability다. 모든 methods가 한 번은 Hit90에 도달하지만, several rows collapse late in training. SABER는 HER보다 높은 mean final success와 AUC를 보이지만, SABER–HER

Table 12: FPP SAC SABER sample-level diagnostic over 1M steps and seeds 1–5. SAC uses twin critics; $|\delta|$ reports the mean absolute residual across the two critics. Values are means and across-seed standard deviations of per-seed diagnostic means.

Metric	Mean ± seed std	Interpretation
Support-positive rate	0.197 ± 0.011	about 20% of batch samples receive high raw weight
$\mathrm{corr}(\psi_i, \lvert\delta_i\rvert)$, Pearson	0.143 ± 0.011	weak positive relation with TD-error magnitude
$\mathrm{corr}(\psi_i, \lvert\delta_i\rvert)$, Spearman	0.209 ± 0.008	weak-to-moderate rank relation
Top-10% TD-error support overlap	0.342 ± 0.014	enriched over the 0.197 support base rate
Top-10% TD-error enrichment	1.83 ± 0.08	support-positive samples appear 1.83x more often in top-error set
Top-20% TD-error support overlap	0.330 ± 0.015	enriched but not equivalent to TD-error selection
$\lvert\delta\rvert_{\mathrm{before}}$, support-positive	0.624 ± 0.043	larger pre-update residual than support-negative group
$\lvert\delta\rvert_{\mathrm{before}}$, support-negative	0.358 ± 0.020	lower mean residual
$\Delta\lvert\delta\rvert$, support-positive	0.022 ± 0.004	larger raw residual reduction after weighted update
$\Delta\lvert\delta\rvert$, support-negative	0.006 ± 0.001	smaller raw residual reduction
$\Delta\lvert\delta\rvert/\Omega$, support-positive	0.013 ± 0.002	larger correction remains after weight normalization
$\Delta\lvert\delta\rvert/\Omega$, support-negative	0.007 ± 0.001	lower per-weight correction
Target variance, support-positive	311.9 ± 29.8	support-positive targets are not lower-variance targets
Target variance, support-negative	172.0 ± 15.1	lower target variance in this diagnostic

final delta $+0.240$의 CI $[-0.320, 0.800]$, AUC delta $+0.220$의 CI $[-0.216, 0.656]$가 모두 0을 포함한다. 또한 Shuffled-weight는 final success 0.590으로 SABER와 같고 AUC 0.586으로 SABER의 0.549보다 약간 높으며, SABER–Shuffled AUC delta는 -0.037 with CI $[-0.209, 0.095]$다. 따라서 DrawerClose는 primary result가 아니다. 이 table의 역할은 더 좁다: Random-fixed-count는 final success 0.0과 AUC 0.133으로 collapse하고 SABER–Random-fixed AUC delta $+0.415$ with CI $[0.074, 0.757]$를 보이지만, Shuffled-weight가 SABER를 match한다. 즉 fixed-count random reweighting이 모든 manipulation settings에서 SABER를 대체하지는 못하는 반면, generic shuffled reweighting can still be competitive in this high-variance stress-test. DrawerClose는 support-conditioned mechanism proof가 아니라 mixed-mechanism limitation evidence다.

Additional DrawerClose DDPG Control Stress-Test

DrawerClose DDPG provides the same caution from a second backbone. SABER improves DDPG+HER in mean final success and AUC, but Random-fixed-count, Shuffled-weight, and PER-style controls remain competitive in at least one aggregate metric. This row is therefore useful as transparency and mixed-mechanism evidence, not as an additional main positive claim.

This DDPG stress-test is directionally consistent with the paper's conservative framing. SABER is better than HER in mean AUC, and it reaches Hit90 earlier among all seeds, but it is not cleanly dominant over controls. Random-fixed-count has the highest mean final success, Shuffled-weight has slightly higher mean AUC than SABER, and PER-style is close in AUC. Thus DrawerClose DDPG reduces the risk of cherry-picking only DrawerOpen while also strengthening the limitation: in high-variance manipulation settings, critic-side reweighting can help, but support-conditioned assignment is not isolated as the unique causal mechanism.

Discussion

What SABER Is

SABER는 HER의 대체물이 아니다. HER는 sparse reward에서 학습 가능한 goal-conditioned targets를 만든다. SABER는 HER가 만든 target 중 어떤 Bellman error를 critic이 더 강하게 fitting할지 정한다.

SABER는 PER의 변형도 아니다. PER는 TD-error priority를 기반으로 sampling distribution을 바꾼다. SABER는 sampling distribution을 유지하고, TD-error magnitude가 아니라 local replay-action support를 기준으로 critic loss weight를 조정한다.

SABER는 broad RL-failure diagnosis framework 전체도 아니다. SABER는 practice-oriented diagnostic intuition을 sparse HER actor-critic에 맞게 축소한 critic-side instantiation이다.

Why SABER Is Still Useful

SABER의 가장 중요한 장점은 strong compute/backbone-scaled baseline보다 항상 높은 return을 얻는 데 있지 않다. 특히 REDQ-HER TQB는 exact FPP SAC common grid에서 AUC 0.867, final success 1.000, median Hit90 300k를 보이며, SABER보다 빠른 environment-step learning curve를 제공한다. 그러나 REDQ-HER는 10-critic ensemble과 UTD 4를 사용하는 다른 architecture/compute regime이다. SABER는 HER relabeling, replay sampling, actor objective, backbone Bellman target을 유지하고, 각 HER Bellman residual이 critic loss에 들어가는 강도만 바꾸는 더 좁은 intervention이다.

이 좁은 설계가 SABER의 장점이다. 기존 sparse HER actor-critic에 붙이기 쉽고, goal selection, replay priority, actor regularization, critic architecture를 동시에 바꾸는 방법보다 effect attribution이 깨끗하다. 또한 random-support, random-fixed-count, shuffled-weight, PER-style, high-RR, REDQ-HER controls와 함께 보면 observed gain이 support-conditioned allocation, generic critic-side regularization, TD-error-like prioritization, update scaling, or backbone scaling 중 어디에 가까운지 진단할 수 있다.

Table 13: Compact control-closure summary for the main claim. SABER is evaluated as a support-conditioned adaptive critic objective in the standard one-update-per-step matched-backbone regime, not as proof that exact structured support is the unique causal mechanism or as a replacement for compute/backbone scaling. Detailed stress-test rows are discussed outside this main closure table.

Evidence	Ruled out	Remaining interpretation
FPP DDPG controls: Random-fixed-count has AUC 0.540, close to HER's 0.539, while Random-support has AUC 0.604, close to SABER's 0.609.	Fixed-count random weighting alone does not explain the gain, but FPP DDPG does not isolate exact sample-level support alignment as an independent cause.	Count-preserving randomization can be a strong critic-side regularizer in DDPG.
FPP TD3/SAC controls: TD3 random-support/fixed-count stay near HER AUC; on the exact SAC common grid, SABER AUC 0.583 exceeds random-support 0.482 and random-fixed 0.526 despite random-fixed final 1.000.	Random reweighting does not explain all SABER gains across backbones, and final success alone can be misleading.	TD3/SAC provide cleaner random-control separation than DDPG, with AUC as the safer learning-speed metric.
High replay-ratio HER and REDQ-HER: DDPG/TD3 $4\times$ match SABER's AUC, SAC $4\times$ exceeds SABER's AUC, and REDQ-HER TQB is strongest in FPP AUC, under larger update/backbone budgets and longer recorded runtimes.	SABER does not dominate compute/backbone scaling in environment-step AUC.	SABER's scope is standard one-update-per-step critic allocation; high-RR and REDQ-HER rows are scaling references.
FPP DDPG/TD3/SAC diagnostics show weak TD-error correlation, high-error enrichment, and larger immediate residual correction in TD3/SAC.	SABER is not identical to TD-error ranking or PER-style largest-error selection.	The strongest mechanism evidence is support-conditioned error allocation and correction, not lower target variance.

따라서 SABER의 publishable contribution은 "REDQ-HER를 대체하는 새 SOTA algorithm"이 아니라, HER 이후 critic fitting pressure를 어디에 배정할지 다루는 lightweight, modular, and interpretable objective라는 점이다. FPP across-backbone evidence와 DrawerOpen DDPG large-effect case는 이 좁은 intervention이 실제 성능과 seed consistency를 개선할 수 있음을 보이고, controls/diagnostics는 이 효과의 조건과 한계를 드러낸다.

Why Critic-Only

Critic-only design은 논문적으로 중요하다. Actor loss까지 support weight를 적용하면, support signal이 critic fitting 효과를 반영하는지 actor objective shaping을 반영하는지 분리하기 어렵다. Critic-only SABER는 actor objective와 replay sampling을 baseline과 동일하게 유지하므로, support signal의 역할을 Bellman regression 쪽으로 좁혀 해석하기 쉽다.

Runtime Cost

SABER는 local support lookup 때문에 runtime overhead가 있다. FPP에서는 performance gain과 함께 moderate slowdown이 나타났고, FetchSlide에서는 slowdown이 훨씬 크다. 따라서 SABER를 항상 더 efficient하다고 주장하면 안 된다.

정확한 claim은 다음이다.

> SABER can improve sample efficiency, but its current local support implementation is not always wall-clock efficient.

Existing SABER summaries include coarse per-update timing fields for the final training state. Table 20 reports their seed-averaged decomposition for rows where the fields are available. These numbers are not a full profiler trace and do not include all Python overhead, environment stepping, evaluation, logging, or hardware-normalized effects. They are useful only for identifying the dominant training-time support overhead in the current implementation.

The decomposition suggests that local context lookup and support scanning, rather than the scalar weight normalization itself, dominate the extra per-update cost. This motivates cached or approximate support lookup as the main engineering path for future work. Because old artifacts do not store a uniform full profiler trace, Table 20 is used as overhead diagnosis, not as a controlled latency benchmark.

High replay-ratio HER and REDQ-HER baselines make this tradeoff sharper. In FPP DDPG, HER $4\times$ reaches a similar AUC to SABER, but requires 4x gradient updates and about 16.1 recorded training hours, compared with SABER's one-update-per-step run and about 6.7 hours. In FPP TD3, HER $4\times$ also matches SABER's AUC within the paired CI, but has lower final success, fewer Hit90 seeds, and longer runtime. On the exact FPP SAC common grid, HER $4\times$ reaches AUC 0.768 versus SABER's 0.583, but takes about 47.0 recorded training hours versus 9.6 hours. REDQ-HER TQB is stronger still, reaching AUC 0.867, final success 1.000, and median Hit90 300k, but it uses a 10-critic ensemble, UTD 4, target-Q bound, about 89.5 recorded training hours, and only 3.10 observed steps/sec. DDPG HER $8\times$ is even slower and less stable. Therefore runtime discussion compares environment-step AUC, final success/seed consistency, and observed wall-clock/update/backbone budget separately; it does not use AUC/hour as a normalized

Table 14: Meta-World DrawerOpen DDPG large-effect result at 2M steps. The HGR-style sampler-only row changes episode and future-goal sampling using TD-error ranking while keeping loss weights uniform; it is not canonical HGR because importance-sampling correction is disabled. Random-support preserves the per-batch support-positive count but randomly assigns high weights. PER-style uses TD-error-based loss weighting. Hit90 reports the median first-hit step among reached seeds, with reach count in parentheses.

Method	Final	Best	AUC	Median Hit90 Step	Steps/sec
DDPG+HER	0.390 ± 0.534	0.400 ± 0.548	0.207 ± 0.287	960k (2/5)	41.05 ± 0.33
DDPG+HER+HGR-style sampler	0.400 ± 0.548	0.400 ± 0.548	0.311 ± 0.425	450k (2/5)	34.61 ± 0.06
DDPG+HER+SABER	1.000 ± 0.000	1.000 ± 0.000	0.754 ± 0.044	500k (5/5)	23.33 ± 0.97
DDPG+HER+Random-support	1.000 ± 0.000	1.000 ± 0.000	0.706 ± 0.122	510k (5/5)	22.61 ± 0.61
DDPG+HER+PER-style	0.800 ± 0.447	1.000 ± 0.000	0.715 ± 0.090	610k (5/5)	39.02 ± 0.38
DDPG+HER+Random-fixed-count	0.800 ± 0.447	0.800 ± 0.447	0.621 ± 0.352	400k (4/5)	41.00 ± 0.37
DDPG+HER+Shuffled-weight	0.400 ± 0.548	0.420 ± 0.531	0.400 ± 0.547	1735k (2/5)	141.62 ± 53.69

Table 15: FetchSlide agent-family results at 1M environment steps. Values are regenerated from the result pipeline source of truth, `analysis/result_pipeline/output/aggregate_metrics.csv`. Steps/sec is observed throughput and is not a controlled hardware-normalized comparison.

Backbone	Method	Final Success	Best Success	Success AUC	Last-Quarter AUC	Steps/sec
DDPG	HER	0.630 ± 0.104	0.720 ± 0.057	0.324 ± 0.023	0.536 ± 0.059	65.13 ± 0.14
DDPG	HER+SABER	0.590 ± 0.114	0.730 ± 0.045	0.301 ± 0.028	0.463 ± 0.049	22.90 ± 0.20
TD3	HER	0.630 ± 0.182	0.770 ± 0.125	0.281 ± 0.066	0.594 ± 0.121	111.43 ± 0.32
TD3	HER+SABER	0.730 ± 0.172	0.880 ± 0.057	0.337 ± 0.012	0.677 ± 0.054	40.24 ± 0.20
SAC	HER	0.820 ± 0.115	0.950 ± 0.050	0.500 ± 0.030	0.775 ± 0.049	66.78 ± 0.07
SAC	HER+SABER	0.840 ± 0.082	0.950 ± 0.035	0.535 ± 0.022	0.806 ± 0.025	30.57 ± 0.19

efficiency claim. SABER is not a wall-clock SOTA claim and does not dominate compute/backbone-scaled HER in environment-step AUC.

Limitations

본 논문의 주요 limitation은 다음이다. 첫째, SABER는 모든 task/backbone에서 개선되지 않는다. FetchSlide DDPG에서는 HER baseline보다 낮은 final success와 late success를 보였다. 둘째, support predicate는 task-specific context design에 의존하고, 현재 main predicate는 local pool count와 action-bin membership을 보지만 unique action-bin diversity나 density를 직접 강제하지 않는다. 셋째, local support lookup 때문에 runtime overhead가 크며, 현재 구현을 wall-clock efficient method로 주장할 수 없다. 또한 FPP DDPG와 TD3 high replay-ratio $4\times$가 SABER와 comparable AUC를 얻고, FPP SAC high replay-ratio $4\times$와 REDQ-HER TQB는 SABER보다 높은 AUC를 얻기 때문에, SABER가 compute/backbone-heavy scaling보다 environment-step AUC에서 우월하다고 주장할 수 없다. 다만 DDPG/SAC high-RR $4\times$와 REDQ-HER는 훨씬 많은 updates/runtime 또는 larger critic ensemble을 요구하고, TD3 high-RR $4\times$는 final success와 Hit90 seed count가 낮으므로, AUC, final success/seed consistency, update budget, and critic architecture를 분리해서 해석해야 한다. 넷째, FPP DDPG Random-Support, FPP Shuffled-Weight, FPP SAC Random-Fixed-Count, DrawerOpen DDPG Random-Support/PER-Style/Random-Fixed-Count, DrawerOpen TD3 Random-Support, DrawerClose DDPG Random-Fixed/Shuffled/PER-Style, and DrawerClose TD3 Shuffled-Weight 결과는 structured support alignment가 유일한 원인이라는 강한 claim을 배제한다. 반대로 FPP TD3 controls에서는 random-support와 random-fixed-count가 HER 수준에 머물러 SABER의 AUC gain을 설명하지 못하고, FPP SAC에서는 random-fixed-count가 final success를 포화시키지만 SABER보다 낮은 AUC를 보인다. DrawerClose TD3에서는 Random-Fixed-Count가 final collapse를 보지만 Shuffled-Weight가 SABER와 comparable하고, DrawerClose DDPG에서는 Random-Fixed-Count와 PER-style이 final success에서 SABER보다 높으며 Shuffled-weight가 AUC에서 SABER와 comparable하다. 따라서 현재 가장 방어 가능한 해석은 support-conditioned critic reweighting과 generic reweighting regularization이 task/backbone에 따라 다른 비중으로 함께 작동한다는 것이다. 다섯째, FPP DDPG/TD3/SAC sample-level diagnostics는 support signal이 TD-error와 동일하지 않음을 보이고, TD3와 SAC에서는 top-error enrichment와 larger residual correction도 관찰된다. 그러나 support-positive target variance가 더 낮지는 않으므로, diagnostics는 PER-style equivalence를 약화하고 critic-side allocation 해석을 강화하지만 pure lower-noise proof는 아니다.

Conclusion

본 논문은 sparse-reward HER actor-critic에서 relabeled Bellman target의 quantity와 current-policy action region에서의 critic-fit usefulness를 구분해야 한다는 문제를 제기했다. HER는 실패 경험을 relabel하여 학습 가능한 goal-conditioned targets를 늘리지만, 모든 relabeled Bellman residual이 current policy action 근방에서 동일하게 유용한 critic supervision이라고 보장하지는 않는다.

SABER는 이 gap에 대한 좁은 critic-side 접근이다. 각 training sample의 local state-goal-horizon context에서 cur-

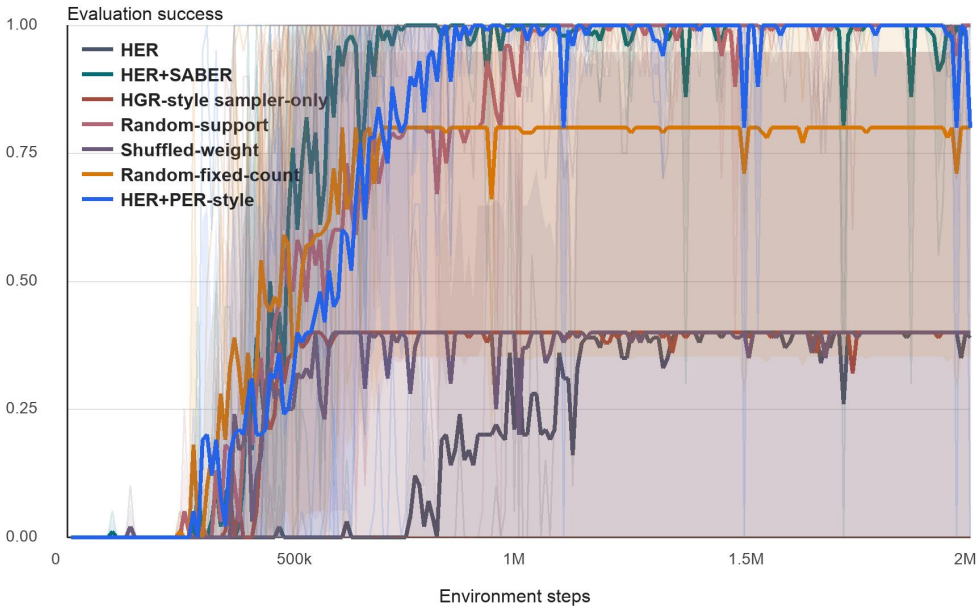


Figure 6: Meta-World DrawerOpen DDPG learning curves for HER, HGR-style sampler-only, HER+SABER, Random-support, PER-style, Random-fixed-count, and Shuffled-weight. Thin lines show individual seeds, and shaded regions show across-seed standard deviation. The sampler-only row produces fast successful runs but only 2/5 seeds reach Hit90; random-support and PER-style are also strong controls. DrawerOpen therefore supports a seed-consistent large-effect SABER case, not a canonical-HGR or universal-causality claim.

Table 16: Meta-World DrawerOpen TD3 stress-test at 2M steps.

Method	Final	Best	AUC	Hit90 Seeds
HER	0.190 ± 0.425	0.210 ± 0.442	0.127 ± 0.283	1/5
HER+SABER	0.590 ± 0.539	0.620 ± 0.522	0.362 ± 0.338	3/5

rent policy action이 replay-action support를 갖는지 확인하고, support-positive sample의 critic Bellman residual에 더 큰 batch-normalized weight를 부여한다. Main variant는 actor objective와 replay sampling distribution을 바꾸지 않으므로, SABER는 HER replacement나 prioritized replay algorithm이 아니라 critic loss reweighting objective로 해석된다.

실험적으로 SABER는 FPP DDPG/TD3/SAC matched HER baselines에서 mean final success와 mean AUC를 높이고, DrawerOpen DDPG에서 HER 대비 large-effect case를 보인다. Exact SAC common-grid 분석에서는 SABER AUC가 HER의 0.471에서 0.583으로 증가하지만, high-RR $4\times$는 0.768, REDQ-HER TQB는 0.867을 달성한다. 따라서 SABER는 compute/backbone scaling을 대체하거나 지배하는 방법이 아니다. Random/PER controls와 DDPG/TD3/SAC diagnostics도 support-conditioned assignment가 TD-error ranking과 동일하지 않음을 보이는 동시에, generic reweighting이 일부 setting에서 경쟁적임을 보여준다. DrawerOpen의 HGR-style row는 importance correction이 없는 sampler-only control이므로 canonical HGR superiority claim에 사용하지 않는다. FetchSlide DDPG failure와 DrawerClose high-variance controls까지 고려하면, 가장 방어 가능한 결론은 SABER가 standard one-update-per-step sparse HER actor-critic에서 critic fitting pressure를 배정하는 lightweight modular objective이며 그 효과가 task/backbone dependent하다는 것이다.

Appendix A: Reproducibility Details

이 appendix는 기존 artifact와 현재 source checkout 기준으로 실험 protocol을 재구성하기 위한 최소 재현성 정보를 정리한다. 제출용 최종본에서는 영어로 번역하고, main paper page limit에 맞추어 supplementary material로 분리할 수 있다.

A.1 Source of Truth and Artifact Scope

FPP 결과는 anonymized code/data package의 `experiment_FPP` trainer/config와 run-level artifacts를 기준으로 한다. FPP SAC HER/SABER/random/high-RR/REDQ rows의 reported AUC, final, and Hit90은 blind artifact ID `FPP-SAC-COMMON-GRID-50K`의 `experiment_FPP/paper_artifacts/fpp_sac_controls_and_redq_1m/common_grid_50k/`를 source of truth로 사용한다. 이 package는 70개 input file의 SHA256 manifest, exact 50k-grid analyzer, seven CPU-only tests, per-seed/aggregate/paired CSVs, NumPy 2.5.1 dependency pin을 포함한다. Native summary metrics는 수정하지

Table 17: DrawerOpen TD3 actor-scope and random-control results at 2M steps. All rows use TD3+HER; method names indicate the SABER-style weighting scope or control.

Method	Final Success	Best Success	Success AUC	Hit90 Seeds
Critic-only SABER	0.590 ± 0.539	0.620 ± 0.522	0.362 ± 0.338	3/5
Actor-only weighting	0.400 ± 0.548	0.410 ± 0.539	0.249 ± 0.341	2/5
Actor+critic weighting	0.600 ± 0.548	0.610 ± 0.534	0.432 ± 0.395	3/5
Random-support	0.600 ± 0.548	0.600 ± 0.548	0.426 ± 0.392	3/5
Random-fixed-count	0.200 ± 0.447	0.200 ± 0.447	0.144 ± 0.322	1/5
Shuffled-weight	0.000 ± 0.000	0.010 ± 0.022	$5.0\times10^{-5} \pm 1.1\times10^{-4}$	0/5

Table 18: Meta-World DrawerClose TD3 stress-test at 2M steps. Best success and Hit90 show that all methods can transiently reach high success, but final success and best-final drop reveal late-training collapse. Paired SABER–HER CIs include zero, and Shuffled-weight matches SABER on final success while slightly exceeding its mean AUC, so this table is not used as a primary improvement or mechanism claim.

Method	Final	Best	Best–Final Drop	AUC	Hit90
TD3+HER	0.350 ± 0.480	1.000 ± 0.000	0.650 ± 0.480	0.329 ± 0.348	70k (5/5)
TD3+HER+SABER	0.590 ± 0.539	1.000 ± 0.000	0.410 ± 0.539	0.549 ± 0.422	80k (5/5)
TD3+HER+Random-fixed-count	0.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	0.133 ± 0.026	100k (5/5)
TD3+HER+Random-support	0.200 ± 0.447	1.000 ± 0.000	0.800 ± 0.447	0.296 ± 0.356	80k (5/5)
TD3+HER+Shuffled-weight	0.590 ± 0.539	1.000 ± 0.000	0.410 ± 0.539	0.586 ± 0.295	30k (5/5)
TD3+HER+PER-style	0.200 ± 0.447	1.000 ± 0.000	0.800 ± 0.447	0.369 ± 0.275	30k (5/5)

않고 pipeline output의 `native_*`와 provenance columns에 보존한다. FPP SAC diagnostic sample files는 blind artifact ID `FPP-SAC-DIAG-1M`의 `experiment_FPP/runs/fpp_sac_saber_diagnostic_ls512_1m/`를 기준으로 한다.

FPP DDPG internal sensitivity rows use blind artifact IDs `FPP-DDPG-LAMBDA-SCHEDULE`, `FPP-DDPG-QUALITY-FLOOR025`, `FPP-DDPG-ANNEAL-300K`, `FPP-DDPG-POSITIVE-HUBER`, and `FPP-DDPG-POSITIVE-MSE`. FetchSlide 결과는 `experiment_FetchSlide/src/saber_fetch_slide/experiment.py`와 run-level artifacts를 기준으로 한다. DrawerOpen DDPG 결과는 `metaworld_Drawer_Open_DDPG`의 runner scripts와 `results_2m` summaries를 기준으로 한다. HGR-style sampler-only artifact `DRAWEROPEN-DDPG-HGR-SAMPLER-ONLY-2M`은 seed별 summaries, selected metrics, configs, environment snapshots, and logs를 포함하지만 canonical HGR의 importance correction은 포함하지 않는다. DrawerOpen TD3와 DrawerClose DDPG/TD3 rows는 보조 stress-test artifacts를 기준으로 한다. Public repository URLs, user namespaces, and raw commit hashes are intentionally omitted from the review manuscript and should appear only in a non-anonymous camera-ready or release archive.

기존 artifact는 environment ID, threshold, seed mapping, training hyperparameters, runtime seconds, and steps/sec를 상당 부분 저장한다. 그러나 모든 old run이 Python/PyTorch/Gymnasium Robotics/MuJoCo/Meta-World exact package version, CPU model, GPU model, CUDA driver, and host metadata를 완전하게 저장하지는 않는다. 따라서 본 논문은 steps/sec를 hardware-normalized claim으로 사용하지 않고, final rerun에서는 machine metadata를 run config와 summary에 함께 저장해야 한다.

A.2 Environment and Reward Convention

Fetch tasks use Gymnasium Robotics Fetch environments with sparse reward and action range $[-1, 1]^4$. The recorded Fetch snapshots use observation dimension 25, goal dimension 3, and state dimension 28 after concatenating observation and desired goal. DrawerOpen and DrawerClose wrap Meta-World `drawer-open-v3`/`drawer-close-v3` as goal-conditioned environments, using drawer handle position as achieved goal and the environment target position as desired goal.

A.3 HER, Terminal Handling, and Seeds

All main Fetch rows use uniform replay with future-goal HER. The replay buffer samples indices uniformly, and each sample is relabeled with probability 0.8 by choosing a future achieved goal from the same episode. Rewards are recomputed after relabeling using the task's sparse reward function.

The terminal convention is:

$$\begin{aligned} \texttt{success_terminates} &= \texttt{false}, \\ \texttt{timeout_bootstrap} &= \texttt{true}. \end{aligned} \tag{34}$$

Thus HER-created success states are not forced to be terminal in the Bellman target, and time-limit transitions allow bootstrapping.

Unless otherwise stated, reported tables use seeds $1, \ldots, 5$. FetchPickAndPlace and FetchSlide rows run for 1M environment steps; DrawerOpen and DrawerClose stress-test rows use 2M environment steps in the reported table.

Table 19: Meta-World DrawerClose DDPG control stress-test at 2M steps. All rows reach best success 1.0 at least once, so final success and AUC reveal late-training stability differences. SABER improves HER in mean AUC, but Random-fixed-count has higher final success and Shuffled-weight has comparable AUC; this table reinforces the mixed-mechanism interpretation rather than a pure structured-support causal claim.

Method	Final	Best	Best–Final Drop	AUC	Hit90
DDPG+HER	0.160 ± 0.182	1.000 ± 0.000	0.840 ± 0.182	0.251 ± 0.030	70k (5/5)
DDPG+HER+SABER	0.280 ± 0.251	1.000 ± 0.000	0.720 ± 0.251	0.324 ± 0.082	20k (5/5)
DDPG+HER+Random-fixed-count	0.480 ± 0.297	1.000 ± 0.000	0.520 ± 0.297	0.319 ± 0.031	70k (5/5)
DDPG+HER+Random-support	0.240 ± 0.429	1.000 ± 0.000	0.760 ± 0.429	0.277 ± 0.050	40k (5/5)
DDPG+HER+Shuffled-weight	0.070 ± 0.157	1.000 ± 0.000	0.930 ± 0.157	0.328 ± 0.064	100k (5/5)
DDPG+HER+PER-style	0.400 ± 0.166	1.000 ± 0.000	0.600 ± 0.166	0.313 ± 0.098	130k (5/5)

Table 20: Observed final-state timing decomposition from stored SABER `summary.json` fields. Values are seconds per update, averaged over seeds 1–5. Support overhead is local context lookup + numba support scan + weight/candidate bookkeeping. The share is computed against actor-critic update plus these support-overhead fields, so it is not a full wall-clock fraction.

Row	Actor-critic update	Local lookup	Numba scan	Support overhead	Support share
FPP DDPG SABER	0.0071	0.0091	0.0030	0.0122	0.63
FPP TD3 SABER	0.0049	0.0082	0.0028	0.0111	0.69
FPP SAC SABER	0.0123	0.0082	0.0027	0.0111	0.47
DrawerOpen DDPG SABER	0.0108	0.0075	0.0021	0.0097	0.47
DrawerOpen TD3 SABER	0.0081	0.0078	0.0014	0.0094	0.54
DrawerClose TD3 SABER	0.0100	0.0067	0.0017	0.0085	0.46

A.4 Model and Optimization Hyperparameters

DDPG and TD3 use two-hidden-layer MLP actors and critics with hidden size 256, ReLU hidden activations, and tanh actor outputs. The critic input is (o, g, a). SAC uses twin critics and a tanh Gaussian actor. Adam is used for actor and critic optimization.

TD3 uses policy delay 2, target policy noise 0.2, and target noise clip 0.5. SAC uses automatic entropy tuning with initial $\alpha = 0.2$, α learning rate 3×10^{-4}, and target entropy $-|\mathcal{A}|$ when unspecified.

The DrawerOpen HGR-style sampler-only control uses the same DDPG backbone, 2M-step budget, batch size 256, HER ratio 0.8, and one update per environment step as the matched DrawerOpen comparison. Episode priority is the mean valid-transition TD-error priority, and candidate future achieved goals are ranked by DDPG TD-error. The row uses $\alpha_{\text{episode}} = \alpha_{\text{goal}} = 1.0$, priority initialization 1.0, $\epsilon = 10^{-6}$, maximum priority 50, and at most 50 future candidates. Critic/actor loss weights are all one. Canonical HGR instead applies episode and goal importance weights to the critic gradient and reports tuned prioritization/correction parameters; therefore this row is explicitly an ablation rather than canonical HGR.

A.5 SABER Support Settings

All support-lookup rows use 5 bins per action dimension and mixed-radix action-bin IDs. Random-fixed-count controls report the same batch-normalized critic-only weighting interface but do not use support lookup. The legacy config name `M_Q_min` corresponds to M_{avail}; it is not a Q-value threshold. `N_gap_min=128` appears in shared configs from older gap-weight variants but is not part of the main support-positive predicate for support-only SABER rows.

A.6 Runtime and Hardware Metadata

The reported steps/sec values are computed as total environment steps divided by recorded training wall-clock seconds. They include training-time SABER support lookup overhead and may include evaluation/logging overhead depending on the runner. They are reported only as observed throughput from old artifacts, not as a controlled wall-clock benchmark, and they are not inference-time costs.

The run configs record `device=cuda` for the reported GPU runs when available. FetchSlide's runner contains a machine metadata collector that records platform, Python version, PyTorch version, CUDA availability, CUDA version, and CUDA device properties. However, not every old artifact referenced by the current draft contains these fields. FPP and DrawerOpen older artifacts also do not consistently store CPU model, GPU model, CUDA driver, or package versions. Consequently, the current draft reports runtime as an observed throughput metric only and does not claim hardware-normalized wall-clock superiority.

Table 21: Environment and sparse reward conventions.

Task	Environment version	Success condition	Reward convention	Horizon
FetchPickAndPlace	`FetchPickAndPlace-v4`	$\|ag - g\| \leq 0.05$	success 0, failure -1	50
FetchSlide	`FetchSlide-v4`	$\|ag - g\| \leq 0.05$	success 0, failure -1	50
DrawerOpen	`drawer-open-v3`	handle-goal distance ≤ 0.05	success 0, failure -1 wrapper	200
DrawerClose	`drawer-close-v3`	handle-goal distance ≤ 0.05	success 0, failure -1 wrapper	200

Table 22: Seed mapping used by the reported runs.

Field	Value for public seed s
environment seed	s
torch seed	s
numpy seed	s
action-noise seed	s
support-weight seed	$20000 + s$
evaluation seed	$10000 + s$

Table 23: Common optimization settings.

Hyperparameter	FetchPickAndPlace	FetchSlide	DrawerOpen/Close
actor lr / critic lr	3×10^{-4} / 3×10^{-4}	3×10^{-4} / 3×10^{-4}	3×10^{-4} / 3×10^{-4}
γ / τ	0.98 / 0.005	0.98 / 0.005	0.98 / 0.005
batch size	256	256	256
replay capacity	300k	300k	300k
warmup / random start	5k / 10k	5k / 10k	5k / 10k
updates per env step	1; high-RR DDPG uses 4 or 8; high-RR TD3/SAC use 4	1	1
eval episodes	20	20	20
eval frequency	DDPG/TD3 native 10k; SAC source runs use 25k or 10k but reported metrics use exact shared 50k grid	25k	10k

Table 24: SABER support lookup settings for rows referenced in the current draft.

Task / row group	N_{local}	M_{avail}	M_B	`local_search_max`	λ_S
FPP DDPG/TD3/SAC SABER	4	4	16	512	1.0
FetchSlide DDPG SABER	4	4	16	1024	1.0
FetchSlide TD3/SAC SABER	4	4	16	512	1.0
DrawerOpen DDPG SABER	4	4	16	512	1.0
DrawerOpen TD3 SABER/control	4	4	16	512	1.0
DrawerClose DDPG SABER/control	4	4	16	512	1.0
DrawerClose TD3 SABER/control	4	4	16	512	1.0

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